

A candidate bound for the ternary divisor problem

A long-direction fourth-derivative refinement

9 September 2026

Abstract

We give a detailed argument intended to establish the pointwise estimate $\Delta_3(x) \ll_\varepsilon x^{541/1209+\varepsilon}$ for the classical symmetric ternary divisor problem. The argument combines a truncated Voronoi formula, differencing, successive one-dimensional van der Corput transforms on sharp domains, and a fourth-derivative estimate along the longer terminal dual direction. A sublevel decomposition treats zeros of that derivative by using a uniform same-direction fourth/fifth derivative condition. All stationary weights, short dual ranges and integer choices of differencing parameters are retained. Explicit polynomial certificates verify the needed model nondegeneracy, and 51 rational affine identities verify the global parameter inequalities.

Working manuscript. This version integrates the long-direction refinement into the full argument, including the boundary subregion, exact derivative certificates and all global parameter certificates. This revision gives explicit uniform inverse charts, sharp-fiber parameterizations, endpoint accounting, and a separate complexity transfer lemma for the complete exact phase construction.

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1 The proposed result, conventions and external inputs

For $n \in \mathbb{N}$, put

$$d_3(n) = \#\{(a, b, c) \in \mathbb{N}^3 : abc = n\}, \quad \Delta_3(x) = \sum_{n \leq x} d_3(n) - \operatorname{Res}_{s=1} \frac{\zeta(s)^3 x^s}{s}.$$

Throughout, $e(t) = \exp(2\pi it)$ and

$$\theta = \frac{541}{1209}, \quad \rho = \frac{265}{403}, \quad u_0 = \frac{138}{403} = 3\theta - 1. \quad (1.1)$$

These constants satisfy

$$\frac{2}{3} - \frac{\rho}{3} = \theta, \quad \frac{1}{3} + \frac{u_0}{3} = \theta. \quad (1.2)$$

Theorem 1.1 (Proposed bound). *For every $\varepsilon > 0$ and all sufficiently large real x ,*

$$\Delta_3(x) \ll_{\varepsilon} x^{541/1209+\varepsilon}.$$

The proof occupies Sections 2–8 and the two appendices. No unproved spacing or moment estimate is an input.

1.1 Uniformity and scales

All normalized phase variables lie in fixed compact boxes with slightly larger open neighborhoods. When positive variables are used, these neighborhoods stay away from zero. Domains are semialgebraic sets of bounded description complexity: the number and degrees of their defining polynomial conditions are bounded independently of the large parameters. Their real coefficients and sharp endpoints may vary. A fixed number of charts, Boolean operations, projections and rectangular cuts is allowed.

The constants in $\ll$ may depend on these fixed complexity and derivative bounds, and on a small positive auxiliary parameter η , but not on x . All physical scales have at most

polynomial size in x . Fixed powers of $\log x$ will be absorbed into x^η , with η reduced finitely often before the final choice of ε . In intermediate displays, the same symbol x^η may denote such a rescaled loss.

Dyadic lengths and bounding rectangles are understood up to fixed factors. Whenever a one-dimensional theorem requires its length parameter to be at least the actual interval length, take a sufficiently large fixed multiple of the nominal ambient scale. This changes only the comparison constants. In particular, the length of an individual sharp fiber need not be comparable to the ambient scale.

In the global argument we distinguish the physical phase scale from its exact exponent budget:

$$R = 27x, \quad F = (RN)^{1/3} = 3F_0, \quad F_0 = (xN)^{1/3}. \quad (1.3)$$

Exact stationary formulas retain F . Nominal parameter choices use F_0 . Also, Π denotes a nominal product of shift lengths, while $p_{\text{int}} = q_1 q_2$ denotes its integer realization. Only comparability, not equality, will be used between these two products.

1.2 Analytic inputs

We use the following published estimates, in the indicated normalizations.

First, Lemma 3 of Ivić–Zhai [1], specialized to $k = 3$, gives

$$\Delta_3(x) = \frac{x^{1/3}}{\pi\sqrt{3}} \sum_{n \leq Y} d_3(n) n^{-2/3} \cos(6\pi(xn)^{1/3}) + O_\eta(x^{2/3+\eta} Y^{-1/3}), \quad 1 \ll Y \ll x. \quad (1.4)$$

An endpoint convention at an integer x changes the left side by $O(d_3(x)) = O_\eta(x^\eta)$, which is harmless here.

Second, the one-dimensional B transform in Vandehey [2, Theorem 1.1] applies on an interval $[a, b]$ with ambient length $M \geq b - a$ if

$$f'' \asymp T/M^2 > 0, \quad |f'''| \ll T/M^3, \quad |f''''| \ll T/M^4.$$

For a real amplitude g of bounded variation, it replaces the sum by the stationary sum with terms

$$\frac{g(x_r)}{\sqrt{f''(x_r)}} e(f(x_r) - rx_r + 1/8), \quad f'(x_r) = r,$$

and an error

$$\ll (\text{Var}(g) + |g(a)|) \left(\frac{M}{\sqrt{T}} + \log(2 + f'(b) - f'(a)) \right). \quad (1.5)$$

Negative curvature is handled by conjugation. In unweighted applications, changes between open, closed and half-weight endpoint conventions cost $O(1 + M/\sqrt{T})$. We explicitly retain this cost. The statement permits an ambient M larger than the fiber length.

Third, the classical third-derivative estimate, as recorded in Heath-Brown [3, equation (1), $k = 3$], is

$$\sum_{n \in I} e(f(n)) \ll M\lambda^{1/6} + M^{1/2}\lambda^{-1/6} \quad \text{if } |f''| \asymp \lambda, \quad |I| \leq M, \quad M \geq 1. \quad (1.6)$$

The derivative has a fixed sign on each interval under consideration. For shorter fibers one may first use their actual length and then enlarge the bound to M ; isolated points are handled separately when needed.

We also use the fourth-derivative instance of the same classical estimate, Heath-Brown [3, equation (1), $k = 4$]:

$$\sum_{n \in I} e(f(n)) \ll M\lambda^{1/14} + M^{3/4}\lambda^{-1/14}, \quad |f^{(4)}| \asymp \lambda, \quad |I| \leq M, \quad M \geq 1. \quad (1.7)$$

The same endpoint and sign conventions apply. If λ is bounded below by a positive constant, the first term already covers the trivial estimate.

Finally, we use real quantifier elimination with a format bound depending only on the input format; see Basu [4, Section 2]. Lemmas 2.3 and 2.4 below spell out the resulting uniform graph, projection and fiber assertions. These facts do not supply an estimate for lattice points near a curved boundary.

2 Elementary operations and a sharp two-dimensional B inequality

Lemma 2.1 (A inequality). *Let D be a finite lattice set in a rectangle of side lengths $O(M_1), O(M_2)$, where $M_1, M_2 \geq 1$, and let $1 \leq q_j \ll M_j$ be integers. Then*

$$\left| \sum_{z \in D} e(f(z)) \right|^2 \ll \frac{(M_1 M_2)^2}{q_1 q_2} + \frac{M_1 M_2}{q_1 q_2} \sum_{\substack{|r_j| < q_j \\ r \neq 0}} \left| \sum_{z \in D \cap (D-r)} e(f(z+r) - f(z)) \right|. \quad (2.1)$$

The triangular weights may instead be retained before absolute values.

Proof. Extend the original summand by zero. Average over the $q_1 q_2$ shifts in a rectangle and apply Cauchy's inequality on the common enlarged support, which has $O(M_1 M_2)$ points. Expanding the square gives multiplicities $(q_1 - |r_1|)(q_2 - |r_2|)$. The diagonal gives the first term; dividing the off-diagonal multiplicities by $q_1 q_2$ gives weights at most one. No regularity of the boundary is required. If $q_1 = 1$, the first shift coordinate is identically zero. $\square$

Lemma 2.2 (Rectangular Abel summation). *Let a be C^2 on an ambient rectangle with arbitrary positive side lengths M_1, M_2 . For any finite base array supported on D ,*

$$\left| \sum_{z \in D} a(z) e(f(z)) \right| \ll V_{\text{rect}}(a) \sup_J \left| \sum_{z \in D \cap J} e(f(z)) \right|, \quad (2.2)$$

where J ranges over rectangles and

$$V_{\text{rect}}(a) = \|a\|_{\infty} + M_1 \|a_1\|_{\infty} + M_2 \|a_2\|_{\infty} + M_1 M_2 \|a_{12}\|_{\infty}.$$

Fixed coefficients may be retained in the base array on both sides.

Proof. Apply integration by parts in each coordinate to the finite measure defined by the base array. The boundary, edge and interior terms have exactly the four variation costs displayed above. Keep 1_D inside every partial sum. The proof works when a side length is less than one; no extension to a rectangle of side length one is necessary. $\square$

2.1 Uniform formulas, sharp fibers and inverse charts

We use the following precise convention for complexity. A real formula has *format at most B* if it uses at most B free and quantified scalar variables, at most B polynomial conditions and Boolean or quantifier operations, and polynomials of degree at most B . Real coefficients are unrestricted. Parameters, including cut endpoints, are free variables. Descriptions with arbitrary coefficients fit this convention by treating the coefficients as parameters; only finitely many Boolean patterns are possible at a fixed format bound. By quantifier elimination over the reals, a formula of format B has an equivalent quantifier-free formula whose format is at most a number $q(B)$. Only the existence of $q(B)$ is used; see [4, Section 2]. For a function, complexity means the format of its graph. Thus the “algebraic families” below are smooth functions with uniformly semialgebraic graphs, with their real branches included in those graphs.

Lemma 2.3 (Uniform semialgebraic calculus). *Fix a format bound B , a derivative order r , and a number L of operations. Starting with sets and function graphs of format at most B , each of the following operations, and any sequence of at most L of them, produces a graph or set of bounded format, depending only on B, r, L : Boolean operations, coordinate projections, images and preimages under the given maps, affine changes and translations, arithmetic operations on their natural domains, derivatives of order at most r where they exist, and inverse maps restricted to a specified injective branch. Restrictions by derivative inequalities are included. The same conclusion holds for a difference quotient at any nonzero step, uniformly as that step tends to zero, and for its derivative extension at zero. Every one-dimensional fiber of any resulting set has a uniformly bounded number of interval and point components.*

Proof. Composition and images are expressed by existentially quantified graph variables. Addition and multiplication add the equations $w = y_1 + y_2$ and $w = y_1 y_2$; division adds $w y_2 = y_1$, $y_2 \neq 0$. The graph of the inverse of H on a specified set C is precisely

$$\{(y, z) : z \in C, (z, y) \in \text{graph}(H)\}.$$

Injectivity makes this relation single-valued. No choice of an unspecified root is part of this operation.

For completeness, if h is differentiable on an open spatial domain, $v = \partial_i h(z)$ is expressed by the usual limit formula

$$\forall \epsilon > 0 \exists \delta > 0 \forall t \quad 0 < |t| < \delta \implies |h(z + t e_i) - h(z) - t v| < \epsilon |t|,$$

with δ also required to keep the indicated segment in the domain. Replace function values by graph variables. The last inequality is the polynomial inequality $(y_t - y_0 - t v)^2 < \epsilon^2 t^2$; the domain condition also has a fixed formula. Iteration gives every derivative of the fixed order r . For $a \neq 0$, the difference quotient has graph

$$\exists y_0, y_1 : (z, y_0) \in \text{graph}(h), (z + a e_i, y_1) \in \text{graph}(h), \quad a w = y_1 - y_0. \quad (2.3)$$

At $a = 0$ use the derivative graph as a separate parameter piece. A general specified direction is handled by replacing e_i by that direction.

Every construction adds a bounded number of variables, conditions and degrees. Apply q after each of the finitely many operations. This gives a recursive bound depending only on B, r, L , without bounds on coefficient sizes or on nonzero denominators. After fixing all but one spatial coordinate, a quantifier-free description involves at most s univariate polynomials of degrees at most d , for uniformly bounded s, d . Discard the identically zero polynomials.

Their at most sd real roots cut the line into at most $2sd + 1$ points and open intervals on which every condition is constant. This proves the fiber assertion, including degenerate parameter values. $\square$

Lemma 2.4 (Sharp-fiber parameterization and endpoint accounting). *Let $S_\alpha \subset \mathbb{R}^2$ be a bounded semialgebraic family of fixed format; α includes all cut endpoints. For $i = 1, 2$, define its coordinate-fiber interior by*

$$\mathfrak{I}_i S_\alpha = \{z : \exists \delta > 0 \forall t (|t| < \delta \implies z + te_i \in S_\alpha)\}. \quad (2.4)$$

This is a family of bounded format contained in S_α . There is a uniform integer N_ such that every i -fiber of $\mathfrak{I}_i S_\alpha$ is a disjoint union of at most N_* open intervals, and every fiber of $S_\alpha \setminus \mathfrak{I}_i S_\alpha$ contains at most N_* points.*

The interval endpoints can be labeled by a bounded number of semialgebraic functions of α and the other coordinate, on semialgebraic parameter pieces of bounded format. Consequently, if the other coordinate ranges over an interval of length $O(M)$, replacing S_α by $\mathfrak{I}_i S_\alpha$ in a lattice sum with summands of absolute value at most A costs $O(A(1 + M))$.

Proof. Formula (2.4) and Lemma 2.3 give bounded format. Apply the root count in that lemma to a fiber. The interior consists of its open interval components. What remains consists only of included interval endpoints and isolated points, with a uniform bound on their total number. This includes fibers of dimension zero and empty fibers.

Here is an explicit endpoint parameterization. Fiberwise closure is defined by

$$z \in \text{cl}_i S_\alpha \iff \forall \epsilon > 0 \exists t \quad |t| < \epsilon, \quad z + te_i \in S_\alpha.$$

The fiberwise boundary $\text{cl}_i S_\alpha \setminus \mathfrak{I}_i S_\alpha$ is a finite set of uniformly bounded cardinality. Partition the parameter space, including the other coordinate, by that cardinality k . Its ordered elements $b_1 < \dots < b_k$ have graphs defined by membership in this boundary and the assertion that exactly $j - 1$ boundary elements lie below b_j . The latter assertion uses at most N_* quantified variables: list these elements and require every smaller boundary element to be one of them. Thus all these graphs have bounded format.

For adjacent boundary elements, membership of their midpoint decides whether the entire open gap belongs to the fiber. Membership of each b_j decides its endpoint convention. Partition further by these finitely many flags. Because the fibers are bounded, the two exterior gaps are empty. This describes every fiber by ordered open intervals and individually included endpoints. The labels may be discontinuous; the argument neither differentiates them nor assumes continuity in any parameter. Summing the uniform endpoint count over the $O(1 + M)$ integer values of the other coordinate proves the final assertion. $\square$

Lemma 2.5 (Common buffered inverse branches). *Let $K \subset \mathbb{R}^2$ be a fixed compact rectangle, and let Ω contain a fixed neighborhood of K . Suppose $F_0, F \in C^m(\Omega)$, $m \geq 4$, have norms at most M and*

$$|\det \text{Hess } F_0| \geq c, \quad \max(|F_{0,11}|, |F_{0,22}|) \geq c$$

on that neighborhood. There are positive constants r, R, δ_0, b and an integer J , depending only on K, Ω, M, c , with the following properties whenever $\|F - F_0\|_{C^2} \leq \delta_0$.

There are J source rectangles C_j covering K , each contained with a fixed buffer in a ball $B_j = B(z_j, R)$ with $\overline{B_j} \subset \Omega$, and a pivot $i_j \in \{1, 2\}$ on each ball. Both F and F_0 have nonzero Hessian determinant and nonzero i_j -th pure second derivative throughout B_j , with uniform lower bounds and the same respective signs. For $H = F, F_0$ write

$$M_H(z) = (H_i(z), z_{3-i}), \quad N_H(z) = \text{grad } H(z) \quad (i = i_j).$$

Both maps are injective on B_j . There are axis-parallel rectangles $A_j^- \Subset A_j^+$ and $Q_j^- \Subset Q_j^+$, common to F, F_0 , such that both inverse maps are defined on neighborhoods of $\overline{A_j^+}$ and $\overline{Q_j^+}$, respectively, and

$$\begin{aligned} M_H(C_j) &\subset A_j^-, & N_H(C_j) &\subset Q_j^-, \\ N_H^{-1}(Q_j^+) &\subset M_H^{-1}(A_j^+) \subset B_j. \end{aligned} \quad (2.5)$$

All the rectangles have fixed positive side lengths, and each inner-to-outer buffer is at least b . The inverse maps have uniform C^{m-1} bounds. The two Legendre functions, formed using these specified inverse branches, satisfy, for $1 \leq k \leq m-2$,

$$\left\| N_F^{-1} - N_{F_0}^{-1} \right\|_{C^k(Q_j^+)} + \|\mathcal{L}F - \mathcal{L}F_0\|_{C^k(Q_j^+)} \leq C_k \|F - F_0\|_{C^{k+1}(\Omega)}. \quad (2.6)$$

The analogous inverse comparison holds on A_j^+ .

If the phase graphs form a family of bounded format, all the chart choices, inverse graphs and rectangles above can be parameterized with bounded format. The source centers can be fixed in advance. The pivot may depend on the model parameter. No chart depends on a sharp domain, on a cut endpoint, or on a physical aspect ratio.

Proof. Increase $M \geq 1$ to bound the operator norms of all Jacobians and their first derivatives used here. At a source center, the least singular value of $\text{Hess } F_0$ is at least c/M . Choose a pivot with $|F_{0,ii}(z_j)| \geq c$. The absolute determinant of the mixed Jacobian is $|F_{0,ii}|$; its norm is bounded, so it too has a uniform least singular value $\sigma > 0$. By decreasing a fixed $R > 0$, arrange that $B(z_j, R) \subset \Omega$, the selected pure derivative is at least $c/2$ in absolute value there, and both model Jacobians vary from their central values by at most $\sigma/8$. Decrease δ_0 so that the perturbed Jacobians differ from the model Jacobians by at most $\sigma/8$. These choices are uniform in the center and parameter. They also preserve determinant and pivot signs throughout the connected ball.

For either map $H_\nu = M_{F_\nu}$ or N_{F_ν} , where $\nu = 0, 1$ denotes model and exact, put $A = DH_0(z_j)$. Then

$$\|DH_\nu(z) - A\| \leq \sigma/4, \quad \|A^{-1}\| \leq \sigma^{-1} \quad (z \in B_j).$$

Integration on the segment between $z, w \in B_j$ gives

$$|H_\nu(z) - H_\nu(w)| \geq (3\sigma/4)|z - w|. \quad (2.7)$$

In particular both maps are injective on the whole ball. If $|y - H_0(z_j)| \leq \sigma R/8$ and $|H_1(z_j) - H_0(z_j)| \leq \sigma R/8$, the map

$$z \longmapsto z + A^{-1}(y - H_\nu(z))$$

is a contraction of the closed radius- R ball into itself: its Lipschitz constant is at most $1/4$, and its displacement of the center is at most $R/4$. Its unique fixed point lies within $R/3$ of z_j . Thus the two images contain the same target ball of radius $\sigma R/8$, and their inverses there take values strictly inside B_j .

We give the nesting choices to ensure compatibility of the two inverses. First choose concentric mixed rectangles $A^- \Subset A^+$ about $M_{F_0}(z_j)$ whose closures lie strictly inside that common mixed target ball. If A^+ has half-side a , choose the half-side q of Q^+ about $N_{F_0}(z_j)$ sufficiently small in terms of a, M, σ . Indeed (2.7) gives

$$|N_H^{-1}(y) - z_j| \leq \frac{4}{3\sigma}(|y - N_{F_0}(z_j)| + C\delta_0).$$

Consequently

$$|M_H(N_H^{-1}(y)) - M_{F_0}(z_j)| \leq \frac{4M}{3\sigma}(\sqrt{2}q + C\delta_0) + C\delta_0.$$

Choose $q > 0$ so that the first term with $\delta_0 = 0$ is less than $a/4$, and then decrease δ_0 so that the whole expression is less than $a/2$. Choose Q^+ itself strictly within the common gradient target ball, and take Q^- and A^- to have half the respective outer half-sides. Finally choose a source half-side $r > 0$ with $\sqrt{2}Mr + C\delta_0 < \min(a, q)/4$, decreasing δ_0 again if needed. Then $C_j = z_j + [-r, r]^2$ has the asserted image inclusions and fixed buffers. The same estimates hold on slightly larger target rectangles, so inverses are defined on neighborhoods of their closures. A finite r -net of K , fixed before any parameter is chosen, provides the source centers and a uniform number J .

Implicit differentiation of $H_\nu(I_\nu(y)) = y$ first gives $DI_\nu = (DH_\nu \circ I_\nu)^{-1}$. At order k , the term containing $D^k I_\nu$ has coefficient $DH_\nu \circ I_\nu$; every other term is a finite product of derivatives $D^a H_\nu$, $a \leq k$, and lower derivatives of I_ν . This proves the uniform C^{m-1} bounds by induction. For comparison, (2.7) first bounds $|I_1 - I_0|$ by $C \|H_1 - H_0\|_\infty$. Subtract the two differentiated identities and induct. Evaluation of $D^k H_0$ at I_1 rather than I_0 is controlled by its bounded next derivative. For $k \leq m-2$ this gives $\|I_1 - I_0\|_{C^k} \leq C_k \|F - F_0\|_{C^{k+1}}$. The formula $\mathcal{L}H(y) = H(N_H^{-1}(y)) - y \cdot N_H^{-1}(y)$ and $\text{grad } \mathcal{L}H = -N_H^{-1}$ give (2.6).

Finally, use the first pivot when $|F_{0,11}(z_j)| \geq c$, and otherwise the second. This is a fixed semialgebraic rule. Target centers are graph values of the two model maps; all radii are fixed constants. An inverse graph is specified by $z \in B_j$ and $y = M_H(z)$ or $y = N_H(z)$. Lemma 2.3 gives the asserted format bounds, including all parameter-dependent choices. The lower bounds proved above make these graph relations unique branches. $\square$

2.2 The sharp B transformation

For $U, V, P \geq 1$, write

$$\mathcal{R}(U, V, P) = \left(\frac{UV}{\sqrt{P}} + U + V + \sqrt{P} + \frac{UV}{P} \right) \log(2 + U + V + P), \quad (2.8)$$

$$\mathcal{R}_{\text{fine}}(U, V, P) = \left(\frac{UV}{\sqrt{P}} + V + \sqrt{P} + \frac{UV}{P} \right) \log(2 + U + V + P). \quad (2.9)$$

Lemma 2.6 (Sharp two-dimensional B inequality). *Let $f(s, t) = P\mathcal{F}(s/U, t/V)$, with $U, V, P \geq 1$. On a fixed neighborhood of a compact normalized rectangle K , suppose $\mathcal{F}$ is uniformly C^{20} bounded, its graph has bounded format, and*

$$|\det \text{Hess } \mathcal{F}| \geq c > 0, \quad \max(|\mathcal{F}_{11}|, |\mathcal{F}_{22}|) \geq c.$$

Let $D \subset \{(Uz_1, Vz_2) : z \in K\}$ be any semialgebraic sharp domain of bounded format, with arbitrary real parameters and endpoint conventions. Then

$$\left| \sum_{D \cap \mathbb{Z}^2} e(f) \right| \ll \frac{UV}{P} \sup_E \left| \sum_{E \cap \mathbb{Z}^2} e(f_j^*) \right| + \mathcal{R}(U, V, P). \quad (2.10)$$

The supremum includes one of a bounded number of specified source charts j and two arbitrary real rectangles J_1, J_2 . On that chart,

$$\begin{aligned} D_{1,j} &= \mathfrak{I}_1 D^{(j)}, & A_{1,j} &= T_{1,j}(D_{1,j}) \cap J_1, & A_{2,j} &= \mathfrak{I}_2 A_{1,j}, \\ E_{j,J_1,J_2} &= T_{2,j}(A_{2,j}) \cap J_2. \end{aligned} \quad (2.11)$$

Here this notation is for a first-coordinate pivot; the roles of the coordinates are interchanged for the other pivot and the final frequency labels are then restored. The maps are $T_1(s, t) =$

$(f_s(s, t), t)$ and $T_2(\xi, t) = (\xi, g_t(\xi, t))$, with g the mixed phase on the specified inverse branch. The phase $f_j^*(\xi, \zeta) = f(s, t) - \xi s - \zeta t$ uses the gradient inverse on that same chart.

The entire family (2.11), including the free rectangle endpoints, has bounded format. Each E is contained in the gradient image of its source piece and in an inner inverse rectangle of side lengths $O(P/U), O(P/V)$ with a fixed normalized buffer. If $|\mathcal{F}_{11}| \geq c$ throughout, one can always use the order s, t and replace $\mathcal{R}$ by $\mathcal{R}_{\text{fine}}$.

Proof. Apply Lemma 2.5 with $F = F_0 = \mathcal{F}$. Write its normalized source rectangles as $C_1, \dots, C_J$, and their physical versions as $\mathcal{C}_j = \text{diag}(U, V)C_j$. Use the ordered partition

$$D^{(j)} = (D \cap \mathcal{C}_j) \setminus \bigcup_{i < j} \mathcal{C}_i, \quad D = \bigsqcup_{j=1}^J D^{(j)}. \quad (2.12)$$

This assigns every real point, hence every lattice boundary point, exactly once. Lemma 2.3 bounds the format of every piece. Its closure stays inside the larger source ball where both inverse maps and the curvature bounds are valid. We estimate each piece separately.

Fix a first-coordinate pivot and suppress j . The normalized mixed coordinates are $(\xi U/P, t/V)$, while the normalized final coordinates are $(\xi U/P, \zeta V/P)$. The mixed and gradient inverses exist on the full physical rectangles obtained by scaling A^+ and Q^+ from Lemma 2.5. Their branches agree wherever both descriptions apply, by (2.5) and injectivity on the source ball. In particular they agree on every set in (2.11).

Put $D_1 = \mathfrak{I}_1 D$. Lemma 2.4 shows that discarding $D \setminus D_1$ costs $O(1 + V)$. For every integer t , the remaining fiber is a union of at most N_* disjoint open intervals (a, b) . Its closure lies inside the buffered source environment and

$$|f_{ss}| \asymp P/U^2, \quad |f_{sss}| \ll P/U^3, \quad |f_{ssss}| \ll P/U^4.$$

Apply the one-dimensional B theorem to $[a, b]$ with ambient length a fixed multiple of U . Passing from closed or half-weight original endpoints to (a, b) costs at most two original summands. Passing to the strictly interior stationary range $f_s((a, b), t)$ costs at most two stationary summands, each of size $O(U/\sqrt{P})$. Since f_{ss} has constant sign on the entire chart, the stationary point in this interval is unique and all its Maslov factors equal $e(\text{sgn}(f_{ss})/8)$.

Thus, with $L_0 = \log(2 + U + V + P)$, the exact first transformed expression is

$$\sum_{D \cap \mathbb{Z}^2} e(f) = e(\text{sgn}(f_{ss})/8) \frac{U}{\sqrt{P}} \sum_{T_1(D_1) \cap \mathbb{Z}^2} a_1(\xi, t) e(g(\xi, t)) + O(E_1),$$

where

$$E_1 \ll (1 + V)(1 + U/\sqrt{P})L_0. \quad (2.13)$$

This includes the B error, all original endpoint decisions, all dual endpoint decisions, and all isolated original fibers. Disjoint source intervals give disjoint open stationary intervals because T_1 is injective. Hence the displayed dual sum has multiplicity one.

Writing $z = M_{\mathcal{F}}^{-1}(\xi U/P, t/V)$, the amplitude and mixed phase are

$$a_1(\xi, t) = |\mathcal{F}_{11}(z)|^{-1/2}, \quad g(\xi, t) = f(s(\xi, t), t) - \xi s(\xi, t).$$

The amplitude has bounded C^2 norm in normalized mixed coordinates on the full outer rectangle. Lemma 2.2 therefore bounds its weighted sum by a constant times

$$\sup_{J_1} \left| \sum_{A_1 \cap \mathbb{Z}^2} e(g) \right|, \quad A_1 = T_1(D_1) \cap J_1.$$

The cuts may be taken within the mixed ambient rectangle; all their endpoints are free real parameters. The variation bound remains uniform when $P/U < 1$, because the corresponding derivative factor is U/P and the ambient side factor is P/U .

On this same mixed environment, implicit differentiation gives

$$g_{tt} = f_{tt} - \frac{f_{st}^2}{f_{ss}} = \frac{\det \text{Hess } f}{f_{ss}}, \quad |g_{tt}| \asymp P/V^2. \quad (2.14)$$

The third and fourth t derivatives are $O(P/V^3)$ and $O(P/V^4)$: this follows by differentiating the mixed inverse, whose normalized derivatives are uniformly bounded by Lemma 2.5. The sign of g_{tt} is constant because both determinant and pivot signs are constant throughout the source ball.

There are $O(1 + P/U)$ integer values of ξ in the mixed rectangle. For each, replace the t fiber of A_1 by its interior, setting $A_2 = \mathcal{I}_2 A_1$. Its discarded points cost $O(1)$ per ξ . Apply the one-dimensional B theorem on each remaining open interval, as above, now with ambient length a fixed multiple of V . Original endpoint changes cost $O(1)$ and dual endpoint changes $O(V/\sqrt{P})$ per interval. The number of intervals is uniformly bounded by Lemma 2.4, uniformly in J_1 . After multiplying by the preceding Abel factor $O(U/\sqrt{P})$, the total second error is

$$E_2 \ll \frac{U}{\sqrt{P}}(1 + P/U)(1 + V/\sqrt{P})L_0. \quad (2.15)$$

Set $D_2 = T_1^{-1}(A_2)$. The identity $g_t = f_t \circ T_1^{-1}$ gives $T_2 T_1 = \text{grad } f$ and therefore

$$E = T_2(A_2) \cap J_2 = \text{grad } f(D_2) \cap J_2 \subset \text{grad } f(D), \quad D_2 \subset D_1 \subset D. \quad (2.16)$$

The second stationary phase is exactly f_j^* on the original gradient branch. Its amplitude is $V/\sqrt{P}$ times the function

$$a_2(\xi, \zeta) = \left| \frac{\mathcal{F}_{11}}{\det \text{Hess } \mathcal{F}} \right|^{1/2} (N_{\mathcal{F}}^{-1}(\xi U/P, \zeta V/P)).$$

This function is defined and has bounded normalized C^2 norm on the full outer gradient rectangle, before any sharp restriction. A second application of Lemma 2.2 gives the factor $O(UV/P)$ and the second rectangular cut J_2 . No endpoint half-weight remains as a discontinuous amplitude: every such change was already charged in E_1 or E_2 .

Formula (2.11) is a parameterization of all the sets over which we take a supremum. Each operation in it has bounded format by Lemmas 2.3 and 2.4. In particular the fiber interiors are taken as real sets for every real value of the other coordinate, before restricting either coordinate to integers. No set of integers, parameter-dependent enumeration of lattice points, or union over all possible cuts is inserted into these formulas. Lemma 2.5 and (2.16) place every E in the designated inner gradient rectangle, irrespective of both Abel cuts.

Finally, expansion of the two errors gives

$$E_1 + E_2 \ll \left(1 + 2V + \frac{2U}{\sqrt{P}} + \frac{UV}{\sqrt{P}} + \sqrt{P} + \frac{UV}{P} \right) L_0 \ll \mathcal{R}_{\text{fine}}(U, V, P).$$

Here $1 \leq V$ and $U/\sqrt{P} \leq UV/\sqrt{P}$. For the other pivot exchange U, V in this calculation and restore $(\xi, \zeta) = (f_s, f_t)$ at the end; the common remainder is $\mathcal{R}$. Sum over the disjoint chart pieces in (2.12). Their bounded number is absorbed in the constant and their specified branches are included in the supremum. This proves (2.10) with uniform constants for all sharp parameters, endpoint coincidences, thin domains and short dual sides. $\square$

Remark 2.7. The complete trivial dual main term in Lemma 2.6 is

$$\frac{UV}{P}(1 + P/U)(1 + P/V) = P + U + V + UV/P. \quad (2.17)$$

The rectangular hull of a transformed domain is used for counting and amplitude norms only; it does not replace the domain in an oscillating sum. Any later A operation is applied after chart splitting, so that correlations do not mix different inverse branches.

3 The model phase and exact finite differences

For a smooth function with an invertible local gradient, our Legendre convention is

$$(\mathcal{L}G)(p, q) = G(u, v) - pu - qv, \quad (p, q) = \text{grad } G(u, v).$$

In particular $\text{Hess}(\mathcal{L}G) = -(\text{Hess } G)^{-1}$.

Lemma 3.1 (Uniform model nondegeneracy). *Let u, v vary in a fixed positive compact box and let (a, b) vary in a compact set separated from $(0, 0)$. Set*

$$G_0(u, v) = \frac{a\sqrt{u}}{v} + \frac{bu^{3/2}}{v^2}, \quad \Phi_0 = \mathcal{L}G_0, \quad K_0 = \partial_1 \Phi_0, \quad \Psi_0 = \mathcal{L}K_0. \quad (3.1)$$

On local inverse branches, the determinants of $\text{Hess } G_0$ and $\text{Hess } K_0$ are uniformly bounded away from zero. Each of these two Hessians has a uniformly nonzero maximum of its pure second derivatives. Moreover,

$$\max(|\Psi_{0,111}|, |\Psi_{0,222}|) \geq c > 0 \quad (3.2)$$

uniformly on the corresponding compact inverse environments. In addition, the same second coordinate satisfies

$$\max(|\Psi_{0,2222}|, |\Psi_{0,22222}|) \geq c > 0. \quad (3.3)$$

Proof. Put

$$\begin{aligned} D &= a^2v^2 + 4abuv + 6b^2u^2 = (av + 2bu)^2 + 2b^2u^2, \\ E &= a^2v^2 + 6abuv + 18b^2u^2 = (av + 3bu)^2 + 9b^2u^2. \end{aligned}$$

Direct differentiation, with $\text{Hess } K_0$ evaluated at $\text{grad } G_0(u, v)$, gives

$$\det \text{Hess } G_0 = -\frac{3D}{4uv^6}, \quad \det \text{Hess } K_0 = -\frac{160u^2v^{10}E}{27D^3}. \quad (3.4)$$

These quantities are nonzero. The two pure second derivatives of G_0 could vanish simultaneously only if $av = 3bu$ and $av = -3bu$, which is impossible on the parameter set.

The pure second derivatives of K_0 have quartic homogeneous numerators, and the pure third derivatives of Ψ_0 have degree-nine homogeneous numerators. Appendix A gives the full polynomials, the derivative identities producing them, and exact Bézout certificates showing that each pair has no common direction zero. The direction $a = 0$ is checked separately there. For (3.3), the same appendix gives the degree-fourteen and degree-nineteen numerators Q_{14} and Q_{19} , their Bézout identity modulo 101, and the missing projective direction. This proves all the required noncoincidences. Continuity and compactness give uniform positive lower bounds. All normalized upper derivative bounds on buffered inverse charts follow from the same determinant bounds and smoothness. $\square$

3.1 The exact family

For small $e_0 \geq 0$, define

$$\mathcal{F}_{e_0}(u, v) = \frac{u^{3/2}}{v} \left(\frac{3}{c^2 + c + 1} \right)^{3/2}, \quad c = (1 + e_0 u/v)^{1/3}. \quad (3.5)$$

If $\omega = (\omega_1, \omega_2)$ is a unit vector and $\beta > 0$ is small, put

$$G_{e_0, \beta, \omega}(u, v) = \frac{\mathcal{F}_{e_0}(u + \beta\omega_1, v + \beta\omega_2) - \mathcal{F}_{e_0}(u, v)}{\beta}. \quad (3.6)$$

The continuous extension at $\beta = 0$ is understood only for derivative comparison. By the integral representation of a difference quotient,

$$\|G_{e_0, \beta, \omega} - G_0\|_{C^k} \ll e_0 + \beta, \quad a = \frac{3}{2}\omega_1, \quad b = -\omega_2, \quad (3.7)$$

for every fixed k , on a slightly enlarged positive compact set. Indeed $\mathcal{F}_0 = u^{3/2}/v$, and its directional derivative is precisely (3.1). The direction parameters form a fixed ellipse separated from the origin. The constants in (3.7) include the axis directions. The radical family has uniformly bounded derivatives of every fixed order in this neighborhood.

Lemma 3.2 (Compatible inverse charts for the second difference). *Let $G = G_{e_0, \beta, \omega}$, with ω a unit vector, and put $\Phi = \mathcal{L}G$ using the branches specified below. There are a uniformly bounded number of source charts and common exact/model inverse rectangles $Q^- \Subset Q^\circ \Subset Q^+$, with uniform positive buffers, such that on the common outer rectangles*

$$\|\Phi - \Phi_0\|_{C^{13}(Q^+)} \ll e_0 + \beta.$$

Every first-B domain, including both Abel cuts, belongs to its designated Q^- in normalized coordinates. There is a fixed $c_0 > 0$ such that for $0 < \lambda \leq c_0$ the phase

$$\Phi_\lambda(p, q) = \frac{\Phi(p + \lambda, q) - \Phi(p, q)}{\lambda} = \int_0^1 \Phi_p(p + t\lambda, q) dt \quad (3.8)$$

is defined on a neighborhood of $\overline{Q^\circ}$ and satisfies

$$\|\Phi_\lambda - K_0\|_{C^9(Q^\circ)} \ll e_0 + \beta + \lambda, \quad K_0 = \Phi_{0,p}. \quad (3.9)$$

Its Hessian determinant and pure second maximum are uniformly nonzero for sufficiently small fixed upper bounds on e_0, β, λ . There is a further uniformly bounded chart cover of $\overline{Q^-}$ with common buffered inverse rectangles for $\text{grad } \Phi_\lambda$ and $\text{grad } K_0$. On each such rectangle, the specified terminal Legendre functions obey

$$\|\Psi - \Psi_0\|_{C^5} \ll e_0 + \beta + \lambda, \quad \Psi = \mathcal{L}\Phi_\lambda, \quad \Psi_0 = \mathcal{L}K_0, \quad \max(|\Psi_{2222}|, |\Psi_{22222}|) \geq c > 0.$$

On their respective buffered environments one also has

$$\|G\|_{C^{24}} + \|\Phi\|_{C^{22}} + \|\Phi_\lambda\|_{C^{21}} + \|\Psi\|_{C^{20}} \leq C. \quad (3.10)$$

All constants and chart counts are independent of the small nonzero steps and of ω , including its axis directions. The sharp domain and every cut endpoint are independent free parameters in these assertions.

Proof. Choose nested positive source rectangles in advance, with enough room for the translations in (3.6). The model parameters $(a, b) = (3\omega_1/2, -\omega_2)$ range over a compact ellipse separated from zero. Lemma 3.1 supplies uniform determinant and pivot margins on these rectangles. Also (3.7) gives C^{17} closeness of G to G_0 .

We first justify all upper regularity needed for the inverse estimates. Formula (3.5), with its positive radical branches, has a uniformly bounded C^{25} norm on the larger positive rectangle. The identity

$$G(z) = \int_0^1 \omega \cdot \text{grad } \mathcal{F}_{e_0}(z + t\beta\omega) dt$$

gives a uniform C^{24} bound for G . It also proves all the stated comparison bounds without dividing an error by β . Decrease the fixed upper bounds on e_0, β to preserve the model Hessian margins.

Apply Lemma 2.5 to $F_0 = G_0$ and $F = G$. The common mixed rectangles and gradient rectangles are chosen by that lemma, with centers given by the model maps. They are the rectangles to be used in the first B application: Lemma 2.6 uses their source charts, and (2.16) places every resulting normalized dual domain inside its Q^- . Further subdivisions, if any, use restrictions of these branches; uniqueness on the original source ball ensures agreement. Formula (2.6) with $k = 13$ proves the C^{13} comparison from the available C^{17} bound. The gradient inverse $I = (\text{grad } G)^{-1}$ has a uniform C^{23} bound, because differentiating $\text{grad } G(I(y)) = y$ to order $r \leq 23$ uses derivatives of G of order at most $r + 1 \leq 24$ and the uniformly bounded inverse Hessian. Hence $\Phi = G \circ I - y \cdot I$ has, in particular, a uniform C^{22} bound on a neighborhood of $\overline{Q^+}$.

Here is the buffer and branch argument for the next A operation. Reserve an intermediate rectangle Q° whose distance from Q^- and from the complement of Q^+ is at least a fixed $b_1 > 0$. Choose $c_0 < b_1/4$, and decrease it later if necessary. Then

$$\overline{Q^\circ} + [0, c_0]e_1 \subseteq Q^+. \quad (3.11)$$

The ordered partition (2.12) and the triangle inequality are applied before A. For a single branch contribution with normalized domain $E \subset Q^-$, a positive normalized shift λe_1 has correlation domain

$$E_\lambda = E \cap (E - \lambda e_1). \quad (3.12)$$

Both endpoint phase values are evaluations of this same Φ . For $y \in E_\lambda$, the segment $y + [0, \lambda]e_1$ lies in the convex rectangle Q^- , even if it leaves E . Formula (3.11) additionally defines the same difference quotient on a full neighborhood of $\overline{Q^\circ}$, without any assumption that E is connected or has interior. Thus no continuation through a domain boundary or onto a different inverse branch is used.

For every spatial multi-index $|\alpha| \leq 9$, subtract K_0 in the integral formula to obtain

$$\begin{aligned} \partial^\alpha(\Phi_\lambda - K_0)(y) &= \int_0^1 \partial^\alpha(\Phi_p - \Phi_{0,p})(y + t\lambda e_1) dt \\ &\quad + \int_0^1 [\partial^\alpha \Phi_{0,p}(y + t\lambda e_1) - \partial^\alpha \Phi_{0,p}(y)] dt. \end{aligned}$$

The first integral is $O(e_0 + \beta)$ by the C^{13} comparison; the second is $O(\lambda)$ by the uniform next derivative of Φ_0 . This proves (3.9), with no $(e_0 + \beta)/\lambda$ loss. For $|\alpha| \leq 21$ the same formula, without subtraction, and the C^{22} bound of Φ give the uniform C^{21} bound for Φ_λ on Q° .

Every point of Q° belongs to the chosen model inverse environment of G_0 , with its inverse in a fixed positive compact source set. Lemma 3.1 therefore supplies uniform determinant and pure second margins for K_0 on a neighborhood of $\overline{Q^\circ}$. They transfer to Φ_λ by its C^9

comparison, after decreasing the three small upper bounds. Apply Lemma 2.5 again, now on $\overline{Q^-}$ with ambient domain Q° , to $F_0 = K_0$, $F = \Phi_\lambda$. The number of source centers and their buffers are uniform: apply the geometric construction after translating the first-stage rectangles to fixed rectangles, and translate the source charts back. Their side lengths and buffers are fixed, so a fixed finite grid of offsets from the model centers suffices. Every resulting inverse identity and phase is then written in the original (p, q) coordinates. These second-stage source charts are used in the second B application, again with an ordered partition before estimating any sums.

On the common terminal rectangles, (2.6) with $k = 5$ and the C^9 comparison prove the stated C^5 closeness of Ψ and Ψ_0 . The model fourth/fifth lower bound holds on these whole compact inverse environments by Lemma 3.1. Choose e_0, β, c_0 so that the two derivative errors are each less than half its fixed margin; the same-coordinate maximum for Ψ follows. The derivative budget $C^{17} \rightarrow C^{13} \rightarrow C^9 \rightarrow C^5$ thus has spare derivatives at both inverse steps. Finally the C^{21} bound on Φ_λ and its determinant margin give a C^{20} bound on its gradient inverse. Its Legendre formula gives the required C^{20} bound on Ψ . This proves (3.10).

For either $\mathcal{H} = G$ or $\mathcal{H} = \Phi_\lambda$, restoring the physical scales consequently gives

$$\left| \partial_s^i \partial_t^j [P\mathcal{H}(s/U, t/V)] \right| \leq CPU^{-i} V^{-j}, \quad i + j \leq 20. \quad (3.13)$$

Thus both B applications in Proposition 4.2 have the required upper bounds with constants independent of scale and aspect ratio. The uniform semialgebraic assertions for these exact branches and all subsequent sharp restrictions are verified separately in the next lemma. $\square$

Lemma 3.3 (Complexity transfer for the exact phase construction). *Fix the initial sharp-domain format and the finitely many derivative orders and operations used in this proof. The exact families $\mathcal{F}_{e_0}, G, \Phi, \Phi_\lambda, \Psi$, their spatial derivatives up to every specified order, all mixed and gradient inverse branches in Lemma 3.2, and all transformed domains in Lemma 2.6 have uniformly bounded semialgebraic format. The assertion is joint in the parameters, including $e_0, \beta, \omega, \lambda$, positive physical scales, real rectangle endpoints and derivative thresholds. It remains uniform as β or λ tends to zero through positive values. Their derivative extensions at zero can be included as separate pieces.*

In particular the A correlations, the original stationary restrictions, and every individual terminal derivative layer have uniformly bounded numbers of intervals and points in each coordinate fiber. No parameter lower bound depending on x is required for this complexity assertion.

Proof. An explicit graph description starts the construction. For $u, v > 0$, the equality $w = \mathcal{F}_{e_0}(u, v)$ is equivalent to the existence of $r, c, t > 0$ satisfying

$$r^2 = u, \quad vc^3 = v + e_0u, \quad t^2(c^2 + c + 1)^3 = 27, \quad vw = urt. \quad (3.14)$$

Positivity selects every radical branch uniquely. These are fixed-degree polynomial equations. The shifts add the polynomial relations $u' = u + \beta\omega_1$, $v' = v + \beta\omega_2$ and $\omega_1^2 + \omega_2^2 = 1$. If w_0, w_1 are the two values from (3.14), the graph of G for $\beta > 0$ adds $\beta w = w_1 - w_0$. At $\beta = 0$ it is the directional derivative $\omega \cdot \text{grad } \mathcal{F}_{e_0}$, whose graph is supplied by Lemma 2.3. This proves uniform format without any coefficient bound involving $1/\beta$.

Spatial derivative graphs of G are obtained from that same lemma. For a specified source ball B_j , the graph of its gradient inverse is given by

$$z \in B_j, \quad (p, q) = \text{grad } G(z).$$

Lemma 2.5 proves uniqueness on the whole buffered target rectangle. Adjoining the graph value $w = G(z)$ and the polynomial relation $v = w - pz_1 - qz_2$ gives the graph of Φ . The

mixed inverse is handled by $p = G_1(z)$ and $q = z_2$, or by the specified second pivot. All target rectangles have model graph values as centers and fixed radii, and the pivot selection uses a fixed derivative inequality. They therefore have bounded format as well.

For $\lambda > 0$, the graph of Φ_λ uses two values v_0, v_1 of this same selected Φ and the equation $\lambda w = v_1 - v_0$. At zero use $w = \Phi_p$. Repeating the derivative-graph and specified inverse-graph constructions gives the terminal inverse and Ψ . The integrals in Lemma 3.2 prove analytic norm estimates; the graph argument here uses only the finite difference formulas. Thus it does not require an assertion that integration preserves semialgebraicity.

For domains, an A correlation is the single formula $z \in D, z + r \in D$. Physical rescaling uses polynomial graph relations such as $s = Uz_1, \xi U = Pp$, with positive scales; it does not increase degrees as the scales vary. The two B transformations use exactly the finitely many operations in (2.11), with every rectangle endpoint retained as a free parameter. The initial stationary expressions (5.8)–(5.10) are also compositions of rational operations and positive radicals of fixed degrees, on their stated positive frequency range. Their domain restrictions therefore fit the same calculus. None of these steps inserts arithmetic support into a real domain.

Finally, for example, a derivative layer is described by adjoining a graph variable $w = \Psi_{2222}(y)$ and the inequalities $\delta/2 \leq |w| \leq \delta$, and projecting out w . Any fifth-derivative or sign restriction uses the same rule. The number of dyadic levels may grow, but each level is one member of this fixed family with free parameter δ ; no growing union is used to obtain a component bound. There is a fixed number of graph, projection and Boolean operations along each construction path, and every derivative order is fixed. Lemma 2.3 supplies a single resulting format bound, followed by a single fiber-component bound. It includes coincident endpoints, vanishing polynomial coefficients and the axis directions of ω . This proves all the assertions. $\square$

Remark 3.4. The frequencies retain their original labels: $p = G_u, q = G_v$. Exchanging the order of the two B transforms changes only the order of calculation. The next A shift always acts on p , so its limiting phase is $\Phi_{0,p} = -u$. Replacing it by $\Phi_{0,q}$ would require a different derivative calculation. The actual oscillating phase is never replaced by its model in the sum; derivative closeness is the only approximation used.

4 Two seven-term local estimates

Lemma 4.1 (Fourth-derivative sublevel decomposition). *Let ϕ belong to a uniformly C^5 -bounded semialgebraic family of bounded description complexity on a fixed interval of length $O(1)$, and suppose that*

$$\max(|\phi^{(4)}(t)|, |\phi^{(5)}(t)|) \geq c > 0.$$

For $L \geq 1, P > 0$ and every interval restriction I within that interval,

$$\left| \sum_{n: n/L \in I} e(P\phi(n/L)) \right| \ll 1 + L^{5/7} P^{1/14} + L^{29/28} P^{-1/14}. \quad (4.1)$$

The estimate also holds on a union of a bounded number of intervals and points. Constants are uniform in the family and in all real endpoints.

Proof. The case $L = O(1)$ is trivial. On $|\phi^{(4)}| \geq c/2$, split into its bounded number of interval components and use (1.7) with ambient length L and $\lambda \asymp P/L^4$. Isolated points in this part contribute $O(1)$.

On the remaining set, $|\phi^{(5)}| \geq c$. Decompose it into dyadic layers

$$\delta < |\phi^{(4)}| \leq 2\delta, \quad L^{-1} \lesssim \delta \lesssim 1,$$

and a final set $|\phi^{(4)}| \lesssim L^{-1}$. Each layer has a uniformly bounded number of interval and point components. On each interval the fifth derivative has constant sign. The mean value theorem therefore bounds its normalized length by $O(\delta)$. In the physical variable, the available interval length and the fourth derivative size satisfy

$$M \ll \delta L, \quad \lambda \asymp P\delta/L^4.$$

Here $\delta L \gtrsim 1$, so a suitable fixed multiple of δL also bounds the number of integer points. Applying (1.7) gives

$$\delta^{15/14} L^{5/7} P^{1/14} + \delta^{19/28} L^{29/28} P^{-1/14}. \quad (4.2)$$

Both powers of δ are positive, and the dyadic sums converge. Isolated points in such a layer can be included in the same bound: the sum of the two terms in the fourth-derivative estimate is at least $2M^{7/8}$ for $M \geq 1$, by the arithmetic-geometric mean inequality.

Each component of the final set has normalized length $O(L^{-1})$, again by $|\phi^{(5)}| \geq c$ and the mean value theorem. It contributes $O(1)$ points, and there are $O(1)$ components. Semialgebraic finiteness provides all the component bounds uniformly, also after intersecting with I . This proves (4.1). $\square$

Proposition 4.2. *Let $1 \leq H \leq X$, let $e_0 = H/X$ be sufficiently small, and let $\beta > 0$ be sufficiently small. Let D be a permitted sharp domain in an H by X box whose normalized coordinates lie in the fixed positive compact set. For $Z > 0$ put*

$$S_5 = \sum_{(h,n) \in D \cap \mathbb{Z}^2} e(ZG_{e_0, \beta, \omega}(h/H, n/X)).$$

Uniformly over unit directions ω , domains and sharp endpoints, both of the following estimates hold:

$$|S_5| \ll x^\eta \left(Z^{13/25} H^{1/2} X^{1/5} + Z^{89/200} H^{1/2} X^{13/40} + \sqrt{XZ} \right. \\ \left. + Z^{6/25} H^{1/2} X^{2/5} + ZH^{-1/4} + X + HX/\sqrt{Z} \right), \quad (4.3)$$

$$|S_5| \ll x^\eta \left(Z^{13/25} H^{21/50} X^{7/25} + Z^{89/200} H^{101/175} X^{347/1400} + \sqrt{XZ} \right. \\ \left. + Z^{6/25} H^{27/50} X^{9/25} + ZH^{-1/4} + X + HX/\sqrt{Z} \right). \quad (4.4)$$

We call these the long-direction and conservative bounds, respectively. The scales have at most polynomial size in x ; equivalently one may retain the fixed logarithmic losses in the proof.

For later reference, their powers (a_j^, b_j^*, d_j^*) in $Z^{a_j^*} H^{b_j^*} X^{d_j^*}$ are*

j	$\star = L$			$\star = S$		
	a_j^L	b_j^L	d_j^L	a_j^S	b_j^S	d_j^S
1	13/25	1/2	1/5	13/25	21/50	7/25
2	89/200	1/2	13/40	89/200	101/175	347/1400
3	1/2	0	1/2	1/2	0	1/2
4	6/25	1/2	2/5	6/25	27/50	9/25
5	1	-1/4	0	1	-1/4	0
6	0	0	1	0	0	1
7	-1/2	1	1	-1/2	1	1

(4.5)

Proof. If $Z < 1$, the last term bounds the trivial $O(HX)$ count. If $1 \leq Z < X$, Lemma 2.6 and (2.17), with lengths H, X , give the trivial dual main term $Z + H + X + HX/Z$. Together with $\mathcal{R}(H, X, Z)$, this is bounded by

$$(X + \sqrt{XZ} + HX/\sqrt{Z}) \log(2 + H + X + Z).$$

This treats every short initial dual range.

Suppose $Z \geq X$, and set

$$U = Z/H, \quad V = Z/X, \quad W = HX/Z, \quad U \geq V \geq 1. \quad (4.6)$$

The first two-dimensional B inequality gives W times a maximal dual sum S_6 , plus $\mathcal{R}(H, X, Z)$. Fix one chart and one actual dual domain E before applying A. Its exact phase is $Z\Phi(k/U, l/V)$, with the frequency labels of Remark 3.4. A positive integer shift τ in the first coordinate has exact correlation phase

$$Z(\Phi((k + \tau)/U, l/V) - \Phi(k/U, l/V)) = P\Phi_\lambda(k/U, l/V), \quad P = H\tau, \quad \lambda = \tau/U.$$

Lemma 3.2 applies when $\lambda \leq c_0$.

Let $T(\tau)$ denote this unweighted correlation sum. Apply the sharp B inequality again, with lengths U, V and parameter P . If $P \geq U$, both terminal dual lengths

$$R_1 = P/U, \quad R_2 = P/V, \quad R_1 \leq R_2$$

are at least one. The exact terminal phase is $P\Psi(\xi/R_1, \zeta/R_2)$, where $\Psi = \mathcal{L}\Phi_\lambda$. For each fixed first coordinate, the C^5 bounds and the same-coordinate fourth/fifth derivative condition in Lemma 3.2 permit Lemma 4.1 along the second coordinate. Lemma 3.3 gives a uniform component bound for every actual terminal fiber and every derivative layer. Consequently its sum is bounded by

$$1 + R_2^{5/7} P^{1/14} + R_2^{29/28} P^{-1/14}.$$

Counting the $O(R_1)$ first-coordinate frequencies and multiplying by UV/P gives

$$V + P^{15/14} R_2^{-2/7} + P^{13/14} R_2^{1/28} = V + P^{11/14} V^{2/7} + P^{27/28} V^{-1/28}.$$

The V term is already included in $\mathcal{R}(U, V, P)$. All derivative-level restrictions occur after the last B transform. They only truncate the actual terminal fibers. Smooth amplitudes have already been removed by rectangular Abel summation, so no derivatives of the layer endpoints or additional stationary boundary errors arise.

If $P < U$, use the complete trivial dual expression $P + U + V + UV/P$. Since $P \leq U$, it is covered by $\mathcal{R}(U, V, P)$. Thus in every terminal range,

$$|T(\tau)| \ll x^\eta \left(P^{11/14} V^{2/7} + P^{27/28} V^{-1/28} + \mathcal{R}(U, V, P) \right). \quad (4.7)$$

As $V \leq U$, this also holds if the first $V^{2/7}$ is enlarged to $U^{2/7}$. This gives the conservative alternative without imposing any additional derivative hypothesis.

Choose $\star \in \{L, S\}$, put $B_L = V$, $B_S = U$, and apply the one-axis version of Lemma 2.1 with

$$\begin{aligned} Q_{0,L} &= Z^{24/25} H^{-1} X^{-2/5}, & Q_{0,S} &= Z^{24/25} H^{-21/25} X^{-14/25}, \\ Q &= \max\{1, \lfloor \min(Q_{0,\star}, c_0 U) \rfloor\}. \end{aligned} \quad (4.8)$$

Extend $T(\tau)$ to negative shifts by its sharp-domain correlation definition. Translating the summation variable gives $T(-\tau) = \overline{T(\tau)}$, so both signs cost only a factor of two. For $Q \geq 2$,

every $1 \leq \tau < Q$ satisfies $\tau/U < c_0$. Average positive powers of τ using Q and bound negative powers conservatively by $\tau \geq 1$. Then (4.7) yields

$$|S_6|^2 \ll x^\eta \left\{ \frac{(UV)^2}{Q} + UV \left((HQ)^{11/14} B_\star^{2/7} + (HQ)^{27/28} V^{-1/28} + U + V + \sqrt{HQ} + \frac{UV}{\sqrt{H}} + \frac{UV}{H} \right) \right\}. \quad (4.9)$$

When $Q = 1$, there are no correlations and $|S_6| \ll UV$ suffices. In either branch,

$$\frac{Z}{\sqrt{Q}} \ll \frac{Z}{\sqrt{Q_{0,\star}}} + \sqrt{HZ} \leq \frac{Z}{\sqrt{Q_{0,\star}}} + \sqrt{XZ}. \quad (4.10)$$

For $Q \geq 2$, use comparability of the floor to the minimum. For $Q = 1$, either $Q_{0,\star} < 2$ or $c_0 U < 2$, and one of the right-hand terms bounds Z .

Since $W\sqrt{UV} = \sqrt{HX}$, the diagonal contributes $Z/\sqrt{Q}$. The first off-diagonal term contributes, respectively,

$$\begin{aligned} A_L &= Z^{1/7} H^{25/28} X^{5/14} Q^{11/28}, \\ A_S &= Z^{1/7} H^{3/4} X^{1/2} Q^{11/28}. \end{aligned}$$

The remaining off-diagonal contributions are common to both choices:

$$Z^{-1/56} H^{55/56} X^{29/56} Q^{27/56}, \quad \sqrt{XZ}, \quad \sqrt{HZ}, \quad H^{3/4} X^{1/2} Q^{1/4}, \quad ZH^{-1/4}, \quad ZH^{-1/2}.$$

In the $Q \geq 2$ branch only, replace nonnegative powers of Q by the corresponding powers of $Q_{0,\star}$. The choice $Q_{0,\star}$ balances the diagonal with $A_\star$ and gives the first term of the corresponding bound. The next contributions give the second, third and fourth terms. Since $H \geq 1$, $ZH^{-1/2}$ is covered by $ZH^{-1/4}$. Finally, for $Z \geq X \geq H \geq 1$,

$$\mathcal{R}(H, X, Z) \ll (HX/\sqrt{Z} + X + \sqrt{XZ}) \log(2 + H + X + Z).$$

For $Q = 1$, use only (4.10); no off-diagonal bound is needed. This proves both (4.3) and (4.4) for every $Z > 0$ and every allowed sharp restriction. $\square$

5 The original divisor sum and the Case A reduction

Choose $Y = \lfloor x^\rho \rfloor$ in (1.4). Partition each of the three factors of n into dyadic intervals $[2^j, 2^{j+1})$, $j \geq 0$. In a fixed factor box, designate the largest scale by X , breaking ties by a fixed index order, and call the other scales A, B . Put

$$M = AB, \quad N = ABX = MX.$$

Every nonempty cutoff box has

$$N \leq Y, \quad N^{1/3} \leq X \leq N, \quad N \leq abn < 8N. \quad (5.1)$$

Define

$$a_m = \#\{(a, b) \in I_A \times I_B : ab = m\}, \quad a_m \leq d(m), \quad \sum_{m \in I_M} |a_m|^2 \ll Mx^\eta,$$

where I_M is the full ambient product interval, of length $O(M)$. No pointwise ordering of the factors is imposed.

One-dimensional Abel summation in the product removes its weight $n^{-2/3}$ at cost $O(N^{-2/3})$: the normalized weight has bounded variation on $[N, 8N)$. Thus it suffices to prove, uniformly in real product cutoffs Y_1, Y_2 , that

$$S_1 = \sum_{m \in I_M} a_m \sum_{\substack{n \in I_X \\ Y_1 < mn \leq Y_2}} e((Rmn)^{1/3}), \quad (5.2)$$

$$x^{1/3} N^{-2/3} |S_1| \ll x^{\theta+\eta}. \quad (5.3)$$

The physical scale is (1.3). The number of factor boxes is $O(\log^3 x)$. Boxes with $N \leq x^{u_0}$ are trivial; their contribution is included explicitly in Section 8.

5.1 Cauchy and the signed first A

For the moment let $1 \leq q \ll X$ be an integer. Set

$$z_m(n) = 1_D(m, n) e((Rmn)^{1/3}), \quad D = \{(m, n) \in I_M \times I_X : Y_1 < mn \leq Y_2\},$$

with zero extension in n . Cauchy gives

$$|S_1|^2 \leq \left(\sum_{m \in I_M} |a_m|^2 \right) \sum_{m \in I_M} \left| \sum_n z_m(n) \right|^2. \quad (5.4)$$

The second sum runs over the full m interval, including $a_m = 0$. Restricting it to the arithmetic support of a_m would not justify the next B transform.

There is a common interval of $K_q \ll X + q$ integer points containing the support of $\sum_{r=1}^q z_m(n+r)$ for every m . Define

$$C_m(h) = \sum_n z_m(n+h) \overline{z_m(n)}, \quad C(h) = \sum_{m \in I_M} C_m(h).$$

The exact shift identity and expansion of its squared norm give

$$|S_1|^2 \leq \left(\sum_{m \in I_M} |a_m|^2 \right) \frac{K_q}{q} \left[C(0) + 2\Re \sum_{1 \leq h < q} (1 - h/q) C(h) \right]. \quad (5.5)$$

The bracket is nonnegative, being a sum of squared norms divided by q . Moreover $C(0) \ll N$. On each dyadic interval $I_H = [H, 2H)$, the endpoint size plus variation of $1 - h/q$ on $I_H \cap [1, q)$ is at most two. Abel summation in h therefore gives

$$|S_1|^2 \ll x^\eta \left(\frac{N^2}{q} + \frac{N}{q} \sum_{\substack{H \text{ dyadic} \\ H \leq q}} S_2(H) \right), \quad (5.6)$$

where

$$S_2(H) = \sup_{J \subset I_H \cap [1, q)} \left| \sum_{h \in J} C(h) \right|,$$

and J ranges over real intervals. It is essential that the joint signed h sum is kept until this point; there is no extra factor H from removing the triangular weight. If $q = 1$, the correlation sum is empty.

5.2 The first stationary phase, with its exact weight

Fix $h \asymp H > 0$ and $n \asymp X$, and set

$$d_h(n) = (n+h)^{1/3} - n^{1/3} > 0, \quad T = FH/X, \quad L = T/M = FH/N, \quad \mathcal{A} = M/\sqrt{T}. \quad (5.7)$$

For each such pair, the m phase and domain are

$$g(m) = R^{1/3} d_h(n) m^{1/3},$$

$$I_M \cap (Y_1/n, Y_2/n] \cap (Y_1/(n+h), Y_2/(n+h)], \quad n, n+h \in I_X.$$

The domain is an interval without an arithmetic restriction. The curvature has size T/M^2 and the next two derivatives obey the B hypotheses.

The stationary point, phase and weight are exactly

$$m_\nu = \frac{\sqrt{R} d_h(n)^{3/2}}{(3\nu)^{3/2}}, \quad (5.8)$$

$$f_\nu(h, n) = g(m_\nu) - \nu m_\nu = \frac{2\sqrt{R}}{3\sqrt{3\nu}} d_h(n)^{3/2}, \quad (5.9)$$

$$g''(m_\nu) = -\frac{2\nu}{3m_\nu}, \quad w_\nu = \frac{R^{1/4} d_h(n)^{3/4}}{\sqrt{2} 3^{1/4} \nu^{5/4}}. \quad (5.10)$$

The Maslov factor is the constant $e(-1/8)$. Every stationary frequency is a positive integer with $cL \leq \nu \leq CL$, for fixed $c, C > 0$. Either no integer frequency occurs, or $L \geq 1/C$ and the number of frequencies is $O(L)$. This is why a separate $+1$ in the frequency count causes no loss; the zero frequency is absent.

Original endpoint changes and isolated fibers cost $O(1)$ per (h, n) ; dual endpoint conventions cost $O(\mathcal{A})$ per pair. Charge them before grouping by ν . Together with the B error this gives

$$E_0 \ll (HX + N\sqrt{HX/F}) \log x. \quad (5.11)$$

Thus an endpoint half-weight never becomes a discontinuous coefficient to which smooth Abel summation is applied.

For explicit normalization, write $h = Hu$, $n = Xv$, $\nu = L\ell$, and put

$$e_0 = H/X, \quad c_e = (1 + e_0 u/v)^{1/3}, \quad \kappa_e = \frac{3}{c_e^2 + c_e + 1}.$$

Then

$$d_h(n) = \frac{H}{3X^{2/3}} uv^{-2/3} \kappa_e,$$

$$f_\nu(h, n) = \frac{2}{27} T \ell^{-1/2} \mathcal{F}_{e_0}(u, v) = T_\nu \mathcal{F}_{e_0}(u, v), \quad (5.12)$$

$$T_\nu = \frac{2\sqrt{R} H^{3/2}}{27\sqrt{\nu} X} \asymp FH/X,$$

$$w_\nu = \frac{\mathcal{A}}{3\sqrt{2}} \ell^{-5/4} u^{3/4} v^{-1/2} \kappa_e^{3/4}.$$

Since ℓ stays in a fixed positive compact interval, $w_\nu/\mathcal{A}$ has uniformly bounded normalized C^2 norm on a larger positive rectangle, before imposing any stationary domain or product cutoff.

For a fixed endpoint convention the stationary domain may be written as

$$D_{\nu,J,Y} = \{(h,n) : h \in J, n, n+h \in I_X, m_\nu \in I_M, \\ Y_1 < nm_\nu, (n+h)m_\nu \leq Y_2\}.$$

It has bounded semialgebraic complexity. Lemma 2.2 bounds its weighted sum by $\mathcal{AM}_3(\nu)$, where

$$\mathcal{M}_3(\nu) = \sup_{J,K_h,K_n} \left| \sum_{D_{\nu,J,Y} \cap (K_h \times K_n)} e(f_\nu(h,n)) \right|.$$

All cuts are real intervals. The order of the maxima and frequency sums is

$$\begin{aligned} S_2(H) &\leq \sup_J \sum_\nu \left| \sum_{D_{\nu,J,Y}} w_\nu e(f_\nu) \right| + E_0 \\ &\leq \sum_\nu \sup_J \left| \sum_{D_{\nu,J,Y}} w_\nu e(f_\nu) \right| + E_0 \\ &\ll \mathcal{A} \sum_{\nu \lesssim L} \mathcal{M}_3(\nu) + E_0. \end{aligned}$$

Since $\mathcal{AL} = \sqrt{T}$, we obtain

$$S_2(H) \ll x^\eta \left(\sqrt{FH/X} \sup_\nu \mathcal{M}_3(\nu) + N\sqrt{HX/F} + HX \right). \quad (5.13)$$

An empty frequency supremum contributes zero. Every ν may have its own maximizing rectangles. The bound is uniform in their endpoints, and no regularity of these endpoints as functions of ν is required.

5.3 The second A and the complete shift average

Fix one unweighted sum S_3 occurring in $\mathcal{M}_3(\nu)$. Apply Lemma 2.1 with lengths H, X . In a full two-axis choice suppose

$$q_1 \asymp \sqrt{\Pi H/X}, \quad q_2 \asymp \sqrt{\Pi X/H}, \quad p_{\text{int}} = q_1 q_2 \asymp \Pi, \quad \alpha = \sqrt{\Pi/(HX)}.$$

In an axis choice take $q_1 = 1$, $q_2 \asymp \Pi$ and $\alpha = \Pi/X$. The exact choices and all admissibility conditions are verified in Section 6.

For each nonzero shift $r = (r_h, r_n)$, set

$$\beta = \sqrt{(r_h/H)^2 + (r_n/X)^2}, \quad \omega = \beta^{-1}(r_h/H, r_n/X), \quad Z = T_\nu \beta.$$

Its correlation is an unweighted sum over the exact $D_3 \cap (D_3 - r)$ with phase $ZG_{e_0, \beta, \omega}$. Consequently Proposition 4.2 applies once the smallness of the normalized shifts is established. No arbitrary coefficient extension of that proposition is needed.

For every $a \geq 0$, the average of β^a is $O(\alpha^a)$. The negative-power average must also be kept. Direct summation gives

$$\begin{aligned} \frac{1}{p_{\text{int}}} \sum_{r_n \neq 0} \beta^{-1/2} &\ll \sqrt{X/q_2}, \\ \frac{1}{p_{\text{int}}} \sum_{\substack{r_n=0 \\ r_h \neq 0}} \beta^{-1/2} &\ll \frac{\sqrt{H/q_1}}{q_2}. \end{aligned} \quad (5.14)$$

Indeed use $\beta \geq |r_n|/X$ in the first sum and $\beta \geq |r_h|/H$ in the second, followed by $\sum_{j < q} j^{-1/2} \ll \sqrt{q}$. In the full case both contributions are $O(\alpha^{-1/2})$. If $q_1 = 1$, the second sum is empty; in the axis case the first alone gives the same result. Nonnegative powers follow from $\beta \ll \alpha$ and the $O(p_{\text{int}})$ number of shifts.

Choose either $\star = \text{L}$ or $\star = \text{S}$ uniformly for the fixed global parameters, before averaging the nonzero second shifts. Averaging the seven terms of (4.3) or (4.4) yields

$$|S_3|^2 \ll x^\eta \left(\frac{(HX)^2}{\Pi} + HX \sum_{j=1}^7 \mathcal{J}_j \right), \quad (5.15)$$

where, for the selected row of (4.5),

$$\mathcal{J}_j = \left(\frac{FH\alpha}{X} \right)^{a_j^*} H^{b_j^*} X^{d_j^*}, \quad 1 \leq j \leq 7. \quad (5.16)$$

The comparability $T_\nu \asymp FH/X$ is uniform, including for the negative power. In both alternatives,

$$\mathcal{J}_7 = \frac{HX}{\sqrt{FH\alpha/X}}.$$

It is the average of a retained boundary error. Replacing β by its smallest possible value would not give the stated estimate. The choice between the two bounds may depend on N, H, X , but not on an unaccounted exceptional set of individual shifts.

6 Case A: parameter choices and their certification

Assume

$$x^{u_0} \leq N \leq x^\rho, \quad N^{1/3} \leq X \leq N^{9/20}. \quad (6.1)$$

In this section and Appendix B, the letters u, v, h, f denote logarithmic exponents:

$$N = x^u, \quad X = x^v, \quad H = x^h, \quad F_0 = x^f, \quad f = (1 + u)/3.$$

They are not the normalized positive coordinates used for the local models.

Choose

$$q_* = (N/x^{u_0})^{2/3}, \quad q = \max\{1, \lfloor q_* \rfloor\}, \quad h_{\text{cap}} = \frac{2}{3}(u - u_0). \quad (6.2)$$

Throughout (6.1),

$$v - h_{\text{cap}} = \frac{11}{1209} + \frac{\rho - u}{3} + \left(v - \frac{u}{3} \right). \quad (6.3)$$

Thus $q_*/X \leq x^{-11/1209}$ and the first A is admissible. If $q = 1$, $q_* < 2$ implies $N = O(x^{u_0})$, so the trivial bound already gives (5.3). We henceforth treat the nonempty correlations.

6.1 The complete small- H range

Put

$$H_t = N^{2/3} F_0^{-1/3} X^{-1/3}, \quad h_t = (2u - f - v)/3. \quad (6.4)$$

The following exact identities hold on the whole Case A range:

$$\begin{aligned} h_t &= \frac{1007}{36270} + \frac{73}{180}(u - u_0) + \frac{1}{3} \left(\frac{9u}{20} - v \right), \\ f - h_t - v &= \frac{733}{7254} + \frac{47}{90}(\rho - u) + \frac{2}{3} \left(\frac{9u}{20} - v \right), \\ u - h_t - v &= \frac{224}{1395} + \frac{13}{90}(u - u_0) + \frac{2}{3} \left(\frac{9u}{20} - v \right). \end{aligned} \quad (6.5)$$

In particular H_t grows as a fixed power of x , and $H_t X$ is smaller than both F_0 and N by fixed powers. For $H \leq H_t$, use the trivial bound $\mathcal{M}_3(\nu) \ll HX$ in (5.13). Its main term is

$$\sqrt{FH/X} HX = H^{3/2} \sqrt{FX} \ll N,$$

and both errors in (5.13) are $O(N)$ by (6.5). This includes empty or short frequency sets. If $H_t > q_*$, all correlations are covered. Every remaining scale lies in the exact region

$$u_0 \leq u \leq \rho, \quad u/3 \leq v \leq 9u/20, \quad h_t \leq h \leq h_{\text{cap}}. \quad (6.6)$$

6.2 Nominal and integer second shifts

Set

$$c = 75h + 16v - 26f, \quad \mathfrak{p}_0 = \frac{12h + 79v - 26f}{63}, \quad \mathfrak{p}_b = 11h + 2f + 7v - 8u. \quad (6.7)$$

On $c \leq 0$ take the axis choice

$$\Pi = H^{-1/76} X^{23/19} F_0^{-13/38}, \quad q_1 = 1, \quad q_2 = \max\{1, \lfloor \Pi \rfloor\}, \quad \alpha = \Pi/X. \quad (6.8)$$

On $c \geq 0$ take

$$\begin{aligned} \Pi_0 &= H^{4/21} X^{79/63} F_0^{-26/63}, \quad \Pi_b = H^{11} F_0^2 X^7 N^{-8}, \\ \Pi &= \max(\Pi_0, \Pi_b), \quad r_1 = \sqrt{\Pi H/X}, \quad r_2 = \sqrt{\Pi X/H}, \\ q_j &= \max\{1, \lfloor r_j \rfloor\}, \quad \alpha = \sqrt{\Pi/(HX)}. \end{aligned} \quad (6.9)$$

Use the long-direction bound (4.3) on the axis region and on the full region $\mathfrak{p}_0 \geq \mathfrak{p}_b$. On the remaining full region $\mathfrak{p}_b \geq \mathfrak{p}_0$, use the conservative bound (4.4). Thus the choice of $\star$ in (5.16) is fixed before averaging the second shifts. At a common boundary either of the applicable, certified choices may be used.

This distinction is needed for the secondary term. If the long-direction seven terms were used throughout the boundary region with $\mathfrak{p} = \mathfrak{p}_b$, its second deficit at

$$(u, v, h) = \left(\frac{265}{403}, \frac{477}{1612}, \frac{254}{1209} \right)$$

would be $-4213/1934400$. The conservative choice makes every required deficit nonnegative, as certified in Appendix B.

The lower choice Π_b is required by the last term in (5.16): in the full case its contribution to (5.13) after (5.15) is comparable to

$$H^{11/8} F_0^{1/4} X^{7/8} \Pi^{-1/8}. \quad (6.10)$$

Thus $\Pi \geq \Pi_b$ pays this term at the exponent level. Using only Π_0 would leave an uncovered part of the parameter region.

6.3 What the exact certificates prove

Appendix B gives nonnegative rational combinations for every required affine inequality. The three Case A regions are the axis region, the full region with $\mathfrak{p}_0 \geq \mathfrak{p}_b$, and the full region with $\mathfrak{p}_b \geq \mathfrak{p}_0$. The 44 certificates on these regions prove that every term of (5.15), after taking a square root and multiplying by $\sqrt{FH/X}$, is $O(N)$, and that both errors of (5.13) are $O(N)$. They also prove

$$\Pi \geq x^{1/1000}, \quad \alpha \leq x^{-1/1000}, \quad F_0 H/N \geq x^{1/1000}, \quad H/X \leq x^{-11/1209}. \quad (6.11)$$

The stronger bound on H/X also uses (6.3) and $h \leq h_{\text{cap}}$. These are identities throughout the regions, not numerical tests at selected points.

For clarity, check that the actual integer shifts satisfy all local hypotheses. In the full case,

$$\log_x \sqrt{\Pi_0 H/X} = c/126 \geq 0,$$

so $r_2 \geq r_1 \geq 1$ and

$$r_j/2 \leq q_j \leq r_j, \quad \Pi/4 \leq p_{\text{int}} \leq \Pi. \quad (6.12)$$

The normalized lengths are at most α , which tends to zero, and $q_2 \geq \sqrt{\Pi}/2$ grows. In the axis case $q_2 \asymp \Pi$ for large x , while the only first-coordinate correlation shift is zero. Thus the negative-power averages (5.14) hold for the rounded integers, and the nonnegative-power averages do also. The smallness of H/X and of every occurring β follows from (6.11). The positive frequency scale condition follows from its third inequality. The final local clamp in (4.8) uses the fixed c_0 already allowed in Proposition 4.2.

Consequently (5.15) is legitimately applicable with these choices, including its negative Z power. Replacing F by $3F_0$ changes only fixed constants in every propagated term. The parameter cuts include their boundaries and leave no gap. We have therefore proved

$$S_2(H) \ll Nx^\eta$$

for every initial dyadic correlation scale, using either the small- H argument or the certificates. Equation (5.6) gives $|S_1| \ll Nq^{-1/2}x^\eta$ after absorbing logarithms. Since $q \asymp q_*$ outside the already treated trivial branch, its weighted exponent is

$$\frac{1}{3} + \frac{u}{3} - \frac{h_{\text{cap}}}{2} = \frac{1}{3} + \frac{u_0}{3} = \theta. \quad (6.13)$$

This proves (5.3) throughout Case A, uniformly in all original product cutoffs.

6.4 The balance determining the exponent

The parameter choices can also be read from the active balanced box. On $v = u/3$, balancing the first long-direction monomial with the second A diagonal gives $\mathfrak{p} = \mathfrak{p}_0$. Its propagated exponent in S_2 is

$$\frac{89f + 177h - 16v}{126} = \frac{89 + 73u + 531h}{378}.$$

Equating this to u gives the first-A balance $h_{\text{opt}} = (305u - 89)/531$. After the Voronoi weight, the corresponding box exponent is

$$E_L(u) = \frac{1}{3} + \frac{u}{3} - \frac{h_{\text{opt}}}{2} = \frac{443 + 49u}{1062}.$$

The equality $E_L(\rho) = 2/3 - \rho/3$ gives $\rho = 265/403$ and $\theta = 541/1209$. This explains the constants; the certificates, not this calculation alone, establish the entire range.

7 Case B: longer largest factors

We now assume

$$x^{u_0} \leq N \leq x^\rho, \quad N^{9/20} \leq X \leq N, \quad M = N/X, \quad L = F/X. \quad (7.1)$$

7.1 The first B transform and removal of its weight

Apply the one-dimensional B transform to the n sum in (5.2) before any Cauchy inequality. Its stationary formulas are

$$\begin{aligned} n_l &= \frac{\sqrt{Rm}}{(3l)^{3/2}}, & f_0(m, l) &= \frac{2\sqrt{Rm}}{3\sqrt{3l}}, \\ w(m, l) &= \frac{(Rm)^{1/4}}{\sqrt{2} 3^{1/4} l^{5/4}}. \end{aligned} \quad (7.2)$$

Stationary frequencies are positive and $l \asymp L$. The endpoint and B errors, summed against $|a_m|$, are

$$\ll x^\eta (N/\sqrt{F} + M). \quad (7.3)$$

If $X > F$, the stationary interval contains $O(1)$ integer frequencies or none. Its main term is bounded by the same quantity, so

$$|S_1| \ll x^\eta (N/\sqrt{F} + M) \quad (X > F). \quad (7.4)$$

Suppose $X \leq F$, so $L \geq 1$. At $m = Mu$, $l = Lv$, the exact weight is

$$w(m, l) = \frac{X/\sqrt{F}}{\sqrt{2} 3^{1/4}} u^{1/4} v^{-5/4}.$$

The normalized amplitude has bounded C^2 norm on a positive ambient rectangle. Rectangular Abel removes it, retaining a_m in the base array and retaining the exact stationary domain, including the product cutoffs. For each resulting unweighted-in- l sum, Cauchy over the full m interval removes a_m , and A in l gives unit-coefficient correlations on the intersected and translated domain. The squared scale factors are

$$(X/\sqrt{F})^2 ML = N, \quad (X/\sqrt{F})^2 (ML)^2 = M^2 F.$$

Consequently

$$|S_1|^2 \ll x^\eta \left(\frac{M^2 F}{q} + \frac{N}{q} \sum_{1 \leq h < q} B_2(h) + \frac{N^2}{F} + M^2 \right). \quad (7.5)$$

Each $B_2(h)$ denotes the absolute value of an unweighted correlation, uniformly over the resulting domains. For $q = 1$ the correlations are empty. Whenever $q \geq 2$, impose

$$q \leq \frac{\min(X, L)}{C \log x}, \quad (7.6)$$

where $C \geq 1$ is fixed. In particular $q \leq L$ and h/L is small for every nonempty correlation.

7.2 The exact local correlation and both short-dual cases

The correlation phase is

$$f_1(m, l) = \frac{2\sqrt{Rm}}{3\sqrt{3}}((l+h)^{-1/2} - l^{-1/2}). \quad (7.7)$$

Set $P = hX$ and $\delta = h/L$. Its exact normalized phase divided by P is

$$\Gamma_\delta(u, v) = \frac{2\sqrt{u}}{3\sqrt{3}} \frac{(v+\delta)^{-1/2} - v^{-1/2}}{\delta}.$$

The integral identity

$$\Gamma_\delta(u, v) = -\frac{\sqrt{u}}{3\sqrt{3}} \int_0^1 (v+t\delta)^{-3/2} dt$$

gives uniformly bounded C^{20} norm on a slightly larger positive rectangle, independently of small δ . It also shows, for every fixed k , that this is a C^k perturbation, of size $O(\delta)$, of

$$\Gamma_0(u, v) = -\frac{1}{3\sqrt{3}} u^{1/2} v^{-3/2}.$$

The exact m, m derivative never vanishes: the phase is $\sqrt{m}$ times a strictly negative function of l . The model Hessian determinant is nonzero, hence the exact normalized determinant is uniformly nonzero for sufficiently small δ . We may therefore use the fixed-order $\mathcal{R}_{\text{fine}}(M, L, P)$ of Lemma 2.6.

Solving the two gradient equations of the model monomial gives a Legendre phase

$$C_1 |r|^{-1/4} |s|^{3/4}, \quad C_1 \neq 0,$$

on its fixed opposite-sign frequency quadrant. On normalized compact inverse environments, $|r|, |s|$ stay bounded away from zero. The pure third derivative in s is

$$\frac{15}{64} C_1 \operatorname{sgn}(s) |r|^{-1/4} |s|^{-9/4},$$

which is uniformly nonzero. The common inverse argument of Lemma 3.2 transfers this to the exact phase.

If both dual lengths $P/M, P/L$ are at least one, use (1.6) in the second dual variable and count the first. Exactly as in the proof of (4.7), this gives $\sqrt{L}P^{2/3} + P^{5/6}$. Including the complete finer remainder yields

$$B_2(h) \ll x^\eta \left(\sqrt{L}P^{2/3} + P^{5/6} + \sqrt{P} + \frac{ML}{\sqrt{P}} + L \right). \quad (7.8)$$

Here ML/P was absorbed by $ML/\sqrt{P}$, since $P = hX \geq 1$.

We verify (7.8) also when a dual side is short. If $P < L$, the complete trivial dual main term

$$P + M + L + ML/P$$

and $\mathcal{R}_{\text{fine}}(M, L, P)$ are bounded, up to a logarithm, by $L + ML/\sqrt{P}$. Indeed $P \leq L$, $L \geq \sqrt{P}$, and $M \leq ML/\sqrt{P}$.

The only remaining short-dual case is $L \leq P < M$. Since $P = hX \geq X$, it implies $X^2 < N$. Consequently

$$\frac{M}{L^2} = \frac{NX}{F^2} \ll N^{5/6} x^{-2/3} \leq x^{-287/2418}, \quad \frac{5\rho - 4}{6} = -\frac{287}{2418}. \quad (7.9)$$

For sufficiently large x , we have $P < M < L^2$, so again $L \geq \sqrt{P}$. Both P and M are bounded by $ML/\sqrt{P}$, and the same absorption applies. This argument uses the actual global range (7.1); it is not asserted for arbitrary M, L . These cases exhaust all short duals, including their equality boundaries, and prove (7.8) for every nonempty correlation.

7.3 Summation, caps and integer rounding

The contribution of $ML/\sqrt{hX}$ to (7.5), divided by its diagonal M^2F/q , is $O(\sqrt{q/X})$, hence is absorbed by (7.6). Also $\sqrt{P} \leq P^{5/6}$. The remaining variable terms for $q \geq 2$ are

$$\frac{a}{q} + bq^{2/3} + c_1q^{5/6}, \quad a = \frac{N^2F}{X^2}, \quad b = NF^{1/2}X^{1/6}, \quad c_1 = NX^{5/6}, \quad (7.10)$$

together with $NF/X + N^2/F + M^2$. For $q = 1$, the corresponding estimate is only $a + N^2/F + M^2$, with no correlation terms inserted.

Choose

$$q_b = (a/b)^{3/5}, \quad q_c = (a/c_1)^{6/11}, \quad K = \frac{\min(X, L)}{C \log x}, \quad q = \max\{1, \lfloor \min(q_b, q_c, K) \rfloor\}. \quad (7.11)$$

If the minimum is at least two, q is between half that minimum and the minimum. Hence

$$\begin{aligned} a/q &\ll a^{2/5}b^{3/5} + a^{5/11}c_1^{6/11} + a/K, \\ bq^{2/3} &\leq a^{2/5}b^{3/5}, \quad c_1q^{5/6} \leq a^{5/11}c_1^{6/11}. \end{aligned}$$

The cap term satisfies

$$a/K \ll \log x \left(\frac{N^2}{X} + \frac{N^2F}{X^3} \right).$$

If the minimum is less than two, $q = 1$. The responsible term among $a/q_b, a/q_c, a/K$ is already at least $a/2$, so the same retained bound follows. This includes $K < 1$ without requiring an impossible nonempty-shift cap for $q = 1$.

Substituting the two balances gives the seven terms

$$\begin{aligned} |S_1|^2 &\ll x^\eta \left(\frac{N^2}{X} + \frac{N^2F}{X^3} + N^{7/5}F^{7/10}X^{-7/10} \right. \\ &\quad \left. + N^{16/11}F^{5/11}X^{-5/11} + \frac{NF}{X} + \frac{N^2}{F} + M^2 \right). \end{aligned} \quad (7.12)$$

Write $N = x^u$, $X = x^v$ and $F_0 = x^{(1+u)/3}$ as before. After taking a square root and multiplying by $x^{1/3}N^{-2/3}$, the seven budget exponents are

$$\begin{aligned} B_1 &= \frac{1}{3} + \frac{u}{3} - \frac{v}{2}, & B_2 &= \frac{1}{2} + \frac{u}{2} - \frac{3v}{2}, \\ B_3 &= \frac{9}{20} + \frac{3u}{20} - \frac{7v}{20}, & B_4 &= \frac{9}{22} + \frac{3u}{22} - \frac{5v}{22}, \\ B_5 &= \frac{1}{2} - \frac{v}{2}, & B_6 &= \frac{1+u}{6}, \quad B_7 = \frac{1}{3} + \frac{u}{3} - v. \end{aligned} \quad (7.13)$$

The last seven certificates in Appendix B prove $B_j < \theta$ on $u_0 \leq u \leq \rho$, $9u/20 \leq v \leq u$. The smallest constant margin is $11/241800$. The bound (7.4) uses B_6, B_7 , so it is covered as well. The fixed relation $F = 3F_0$ changes only constants. This proves (5.3) on the whole Case B range.

8 Assembly of the proposed pointwise bound

We complete the argument for Theorem 1.1. A factor box with $N \leq x^{u_0}$ contains $O(N)$ triples, so its contribution after product Abel is

$$O(x^{1/3}N^{1/3}) \ll x^{1/3+u_0/3} = x^\theta.$$

For every larger nonempty box, (5.1) gives exactly $u_0 \leq \log_x N \leq \rho$ and $N^{1/3} \leq X \leq N$. The two cases $X \leq N^{9/20}$ and $X \geq N^{9/20}$ overlap at the boundary and exhaust this interval. Sections 6 and 7 prove (5.3) uniformly in the product cuts needed by Abel summation. Summing the $O(\log^3 x)$ factor boxes bounds the main oscillating sum in (1.4).

All original factor cutoffs, first-A subintervals, stationary domains, rectangular cuts and shifted intersections lie in the uniform domain families already treated. The estimates are for each original real x ; no averaging in x or almost-everywhere assertion has been used. The intermediate endpoint and isolated-fiber costs are those explicitly retained in the B errors. In particular, the cutoff at Y is part of the product-domain estimate, not a discarded summand.

Choose one auxiliary $\eta > 0$ sufficiently small compared with the desired ε . There are finitely many sums and Cauchy or Abel operations, and only fixed logarithmic powers. Thus the total loss has the form $x^{C\eta}(\log x)^C$ for a fixed C . Choose $\eta < \varepsilon/(2C)$ and absorb the logarithm into $x^{\varepsilon/2}$. The positive margins establishing chart admissibility are independent of this choice. Since $Y = \lfloor x^\rho \rfloor \geq x^\rho/2$ for large x , the remaining tail is

$$x^{2/3+\eta}Y^{-1/3} \ll x^{2/3-\rho/3+\eta} = x^{\theta+\eta}.$$

Together with (1.2), this is the proposed bound $\Delta_3(x) \ll_\varepsilon x^{541/1209+\varepsilon}$. Finite smaller x are covered by enlarging the implied constant. $\square$

A Exact model derivative certificates

This appendix supplies the algebraic details used in Lemma 3.1. All derivatives of Legendre phases are evaluated at the successive points corresponding to the original (u, v) . For a polynomial $P(z) = \sum_{j=0}^d c_j z^j$, write

$$P^{\text{hom}}(a, b) = \sum_{j=0}^d c_j a^{d-j} b^j.$$

A.1 The quartic and degree-nine polynomials

Define

$$\begin{aligned} P_4(z) &= 7 + 70z + 252z^2 + 432z^3 + 324z^4, \\ Q_4(z) &= 1 - 8z - 72z^2 - 108z^3 - 108z^4, \\ R_9(z) &= -1 - 17z - 196z^2 - 920z^3 + 468z^4 + 19332z^5 \\ &\quad + 75600z^6 + 124416z^7 + 93312z^8 + 46656z^9, \\ S_9(z) &= 7 + 168z + 1764z^2 + 11010z^3 + 45711z^4 + 131814z^5 \\ &\quad + 265896z^6 + 363528z^7 + 306180z^8 + 122472z^9. \end{aligned}$$

At $u = v = 1$, put

$$D_{ab} = a^2 + 4ab + 6b^2, \quad E_{ab} = a^2 + 6ab + 18b^2.$$

The exact pure derivatives are

$$K_{0,11} = -\frac{16P_4^{\text{hom}}(a, b)}{9D_{ab}^3}, \quad K_{0,22} = \frac{2Q_4^{\text{hom}}(a, b)}{9D_{ab}^3}, \quad (\text{A.1})$$

$$\Psi_{0,111} = \frac{27R_9^{\text{hom}}(a, b)}{1600E_{ab}^3}, \quad \Psi_{0,222} = \frac{27S_9^{\text{hom}}(a, b)}{25E_{ab}^3}. \quad (\text{A.2})$$

For general positive u, v , let $t = u/v$ and $E_0 = a^2 + 6abt + 18b^2t^2$. The same pure third derivatives are

$$\begin{aligned}\Psi_{0,111} &= \frac{27a^9 R_9(bt/a)}{1600v^{13/2}t^{7/2}E_0^3}, \\ \Psi_{0,222} &= \frac{27a^9 S_9(bt/a)}{25v^{13/2}\sqrt{t}E_0^3}.\end{aligned}\tag{A.3}$$

At $a = 0$ these expressions mean their homogeneous polynomial extensions, not division by a .

A.2 A direct differentiation procedure

Let $I = (\text{Hess } G_0)^{-1}$. Since $\Phi_{0,p} = -u$, put

$$\begin{aligned}r = K_{0,p} = -I_{11} &= \frac{8u^{3/2}v^2(av + 3bu)}{3D}, \\ s = K_{0,q} = -I_{12} &= \frac{2\sqrt{u}v^3(av + 6bu)}{3D}.\end{aligned}\tag{A.4}$$

The homogeneity degree of G_0 in (u, v) is $-1/2$. Consequently Φ_0 has degree $1/3$ in (p, q) , and K_0 has degree $-2/3$. Euler's identity therefore gives

$$\Psi_0 = K_0 - pr - qs = \frac{5}{3}K_0 = -\frac{5}{3}u.\tag{A.5}$$

Use the inverse Jacobian of $(u, v) \mapsto (r, s)$ in (A.4) to differentiate (A.5) three times in r or s . This gives (A.3). In particular, this procedure does not assume r or s is nonzero.

All expressions can be checked using rational functions by setting $u = w^2$, so $\partial_u = (2w)^{-1}\partial_w$. Explicitly define

$$D_p = I_{11}\partial_u + I_{21}\partial_v, \quad D_q = I_{12}\partial_u + I_{22}\partial_v.$$

Then $r = D_p(-u)$, $s = D_q(-u)$ and $\text{Hess } K_0$ has entries $D_i D_j(-u)$. If $J = \partial(r, s)/\partial(w, v)$, the columns of J^{-1} give the coefficients of ∂_r, ∂_s as operators in ∂_w, ∂_v . This finite differentiation procedure verifies (3.4), (A.1) and (A.3) without explicitly solving a high-degree inverse.

There is also a tensor check at $(1, 1)$. The jet is

$$\partial_u^j \partial_v^k G_0(1, 1) = a(1/2)_{\underline{j}}(-1)_{\underline{k}} + b(3/2)_{\underline{j}}(-2)_{\underline{k}},$$

where the underline denotes a falling factorial. Write $v_i = Ie_i$ and let the vector B_{ij} have components $G_0'''(e_r, v_i, v_j)$. Differentiating $\text{Hess } \Phi_0 = -I$ gives

$$\begin{aligned}\Phi_{0,ijk} &= G_0'''(v_i, v_j, v_k), \\ \Phi_{0,ijkl} &= G_0''''(v_i, v_j, v_k, v_l) - B_{ij}^\top IB_{kl} - B_{ik}^\top IB_{jl} - B_{il}^\top IB_{jk}.\end{aligned}$$

Thus if $L_{ij} = \Phi_{0,1ij}$ and $J_0 = L^{-1}$, then

$$\Psi_{0,rrr} = K_0'''(J_0 e_r, J_0 e_r, J_0 e_r), \quad K_{0,ijk} = \Phi_{0,1ijk}.$$

Substitution of the displayed jet gives exactly (A.1) and (A.2).

A.3 Noncoincidence and the missing projective direction

For a list $[c_0, \dots, c_d]$, let $[c_0, \dots, c_d](z)$ mean $\sum_{j=0}^d c_j z^j$. The following two polynomial identities hold in $\mathbb{F}_{101}[z]$:

$$[81, 68, 77, 20](z)P_4(z) + [40, 32, 49, 60](z)Q_4(z) = 1, \quad (\text{A.6})$$

$$[8, 56, 98, 2, 87, 80, 95, 42, 3](z)R_9(z) + [59, 11, 89, 19, 55, 63, 66, 100, 71](z)S_9(z) = 1. \quad (\text{A.7})$$

They are checked by polynomial multiplication and coefficient reduction. The leading coefficients of P_4, Q_4, R_9, S_9 modulo 101 are 21, 94, 95, 60, respectively. Thus the degrees survive reduction and the two pairs are coprime over $\mathbb{Q}$ (equivalently, their integer resultants are nonzero; their residues are 97 and 51). Here and below $\mathbb{Q}$ denotes the field of rational numbers.

At $a = 0$, $b \neq 0$, the unit-point values are

$$K_{0,11} = -\frac{8}{3b^2}, \quad K_{0,22} = -\frac{1}{9b^2}, \quad \Psi_{0,111} = \frac{27b^3}{200}, \quad \Psi_{0,222} = \frac{567b^3}{25}. \quad (\text{A.8})$$

This checks the projective direction omitted by setting $z = b/a$.

Finally, checking at $(1, 1)$ loses no points. At any $(u_*, v_*) > 0$, rescale the analytic variables and replace the parameter pair by

$$(a', b') = \left(a\sqrt{u_*}/v_*, \quad bu_*^{3/2}/v_*^2 \right).$$

The pure second derivatives of K_0 acquire the positive scaling factors u_*^3 and $u_*v_*^2$, and the pure third derivatives of Ψ_0 acquire u_*^{-5} and $u_*^{-2}v_*^{-3}$. The transformed parameter pair stays in a compact set separated from zero. Since D_{ab} and E_{ab} are positive away from $(0, 0)$, the polynomial noncoincidence, the explicit axis values and compactness prove all uniform lower bounds asserted in Lemma 3.1.

A.4 Fourth and fifth derivatives in the same coordinate

In addition to the degree-nine certificates above, we require the same-coordinate condition (3.3). Continuing the inverse-Jacobian differentiation in (A.4) in the second terminal coordinate gives, at $u = v = 1$,

$$\Psi_{0,2222} = -\frac{81Q_{14}^{\text{hom}}(a, b)}{250E_{ab}^5}, \quad (\text{A.9})$$

$$\Psi_{0,22222} = \frac{243Q_{19}^{\text{hom}}(a, b)}{1250E_{ab}^7}. \quad (\text{A.10})$$

The following table gives all coefficients in $Q_d(z) = \sum_{i=0}^d c_i z^i$. The homogeneous convention is that at the beginning of Appendix A.

i	$[z^i]Q_{14}$	$[z^i]Q_{19}$
0	119	1309
1	4522	68068
2	77826	1625778
3	824568	24203784
4	6106629	255656181

i	$[z^i]Q_{14}$	$[z^i]Q_{19}$
5	33762834	2057394312
6	144505512	13172633964
7	487756296	68944652064
8	1305974988	299918873502
9	2759927904	1094360860548
10	4528553832	3361429410120
11	5592351456	8683008133680
12	4917915648	18765796869360
13	2764437984	33609519267840
14	753937632	49138093018080
15	0	57316640278464
16	0	51469491145248
17	0	33532209482688
18	0	14167995980544
19	0	2931309513216

For a polynomial specified by its ascending coefficient list, there is the identity

$$A_{18}(z)Q_{14}(z) + B_{13}(z)Q_{19}(z) \equiv 1 \pmod{101}. \quad (\text{A.11})$$

To specify the certificate without a long unbroken coefficient list, write

$$A_{18}(z) = A^{(0)}(z) + z^{10}A^{(1)}(z), \quad B_{13}(z) = B^{(0)}(z) + z^7B^{(1)}(z),$$

where the ascending lists are

$$A^{(0)} = [45, 18, 2, 88, 53, 55, 93, 1, 71, 67],$$

$$A^{(1)} = [100, 76, 17, 21, 72, 87, 0, 57, 6],$$

$$B^{(0)} = [76, 87, 44, 85, 8, 90, 22],$$

$$B^{(1)} = [11, 90, 96, 45, 74, 100, 12].$$

All coefficients are interpreted in $\mathbb{F}_{101}$. Direct multiplication verifies (A.11). The leading coefficients of Q_{14} and Q_{19} reduce to 3 and 49, so their degrees survive reduction. Their resultant is 29 modulo 101. Consequently the two integer polynomials are coprime over $\mathbb{Q}$ and have no common finite complex zero.

At $a = 0$, the omitted projective direction is

$$\Psi_{0,2222} = -\frac{32319}{250}b^4, \quad \Psi_{0,22222} = \frac{581742}{625}b^5, \quad (\text{A.12})$$

which are both nonzero for $b \neq 0$. Under the positive-coordinate rescaling described above, the fourth and fifth derivatives acquire respectively the positive factors $u_*^{-3}v_*^{-4}$ and $u_*^{-4}v_*^{-5}$. The transformed parameters remain in a compact set separated from zero, and $E_{ab} > 0$ there. Compactness gives the uniform lower bound (3.3). The C^5 comparison in Lemma 3.2 transfers it to the exact phase. Thus zeros of its fourth derivative are treated by Lemma 4.1 without replacing the phase in an exponential sum by its model.

B Exact affine certificates for all parameter ranges

All identities in this appendix are over the rational numbers. They can be verified by comparing coefficients of $1, u, v, h$. Every right side is a nonnegative rational combination of generators that are nonnegative on the stated region. No numerical vertex search is needed to use the certificates.

B.1 Deficits and common generators

Retain

$$u_0 = \frac{138}{403}, \quad \rho = \frac{265}{403}, \quad f = \frac{1+u}{3}, \quad h_t = \frac{2u-f-v}{3}, \quad h_{\text{cap}} = \frac{2(u-u_0)}{3},$$

and $c, \mathbf{p}_0, \mathbf{p}_b$ from (6.7). Write $\mathbf{p} = \log_x \Pi$, $\mathbf{a} = \log_x \alpha$ for the nominal, not rounded, scales. In the axis and main full regions use $(a_j, b_j, d_j) = (a_j^L, b_j^L, d_j^L)$ from (4.5); in the boundary full region use the conservative row $\star = \text{S}$.

The propagated deficits for (5.15) followed by (5.13) are

$$\mathcal{D} = u - \frac{f + 3h + v - \mathbf{p}}{2}, \quad (\text{B.1})$$

$$\mathcal{C}_j = u - \frac{f + h - v}{2} - \frac{h + v + a_j(f + h - v + \mathbf{a}) + b_j h + d_j v}{2}. \quad (\text{B.2})$$

In particular, $\mathcal{C}_j$ is a deficit for the contribution to S_2 , not merely for $\mathcal{J}_j$. Define

$$\mathcal{E}_{\text{end}} = u - h - v, \quad \mathcal{E}_{\text{curv}} = \frac{f - h - v}{2}, \quad \mathcal{Q}_1 = \frac{\mathbf{p} + h - v}{2}.$$

The common generators are

$$\begin{aligned} g_0 &= 1, & g_1 &= u - u_0, & g_2 &= \rho - u, \\ g_3 &= v - u/3, & g_4 &= 9u/20 - v, & g_5 &= h - h_t, \\ g_6 &= h_{\text{cap}} - h. \end{aligned}$$

They are nonnegative on (6.6). The extra generators and the selected exponents are specified in each subregion below.

B.2 Case A: the axis region

Here $c \leq 0$ and the long-direction powers are used:

$$g_7 = -c, \quad \mathbf{p} = \frac{-h + 92v - 26f}{76}, \quad \mathbf{a} = \mathbf{p} - v.$$

The 14 certificates are

$$\mathcal{D} = \frac{329}{24180}g_0 + \frac{6}{35}g_3 + \frac{67}{56}g_6 + \frac{11}{2660}g_7 \quad (\text{B.3})$$

$$\mathcal{C}_1 = \frac{329}{24180}g_0 + \frac{6}{35}g_3 + \frac{67}{56}g_6 + \frac{11}{2660}g_7 \quad (\text{B.4})$$

$$\mathcal{C}_2 = \frac{941}{64480}g_0 + \frac{11}{140}g_3 + \frac{521}{448}g_6 + \frac{87}{21280}g_7 \quad (\text{B.5})$$

$$\mathcal{C}_3 = \frac{5675}{251472}g_0 + \frac{5}{182}g_4 + \frac{3287}{2912}g_6 + \frac{87}{55328}g_7 \quad (\text{B.6})$$

$$\mathcal{C}_4 = \frac{4817}{125736}g_0 + \frac{5}{91}g_4 + \frac{1649}{1456}g_6 + \frac{87}{27664}g_7 \quad (\text{B.7})$$

$$\mathcal{C}_5 = \frac{17}{3720}g_0 + \frac{29}{70}g_3 + \frac{143}{112}g_6 + \frac{13}{10640}g_7 \quad (\text{B.8})$$

$$\mathcal{C}_6 = \frac{15403}{502944}g_0 + \frac{25}{52}g_4 + \frac{757}{832}g_6 + \frac{1}{832}g_7 \quad (\text{B.9})$$

$$\mathcal{C}_7 = \frac{11899}{335296}g_0 + \frac{445}{728}g_4 + \frac{9897}{11648}g_6 + \frac{1191}{221312}g_7 \quad (\text{B.10})$$

$$\mathbf{p} - \frac{1}{1000} = \frac{826529}{22971000}g_0 + \frac{15}{19}g_3 + \frac{24}{19}g_5 + \frac{97}{76}g_6 \quad (\text{B.11})$$

$$-\mathbf{a} - \frac{1}{1000} = \frac{132393863}{1079637000}g_0 + \frac{220}{893}g_4 + \frac{96}{893}g_5 + \frac{337}{3572}g_6 \quad (\text{B.12})$$

$$f + h - u - \frac{1}{1000} = \frac{353789}{5211000}g_0 + \frac{80}{193}g_4 + \frac{2912}{1737}g_5 + \frac{47}{5211}g_7 \quad (\text{B.13})$$

$$v - h - \frac{1}{1000} = \frac{4007}{93000}g_0 + \frac{39}{35}g_3 + \frac{13}{28}g_6 + \frac{1}{140}g_7 \quad (\text{B.14})$$

$$\mathcal{E}_{\text{end}} = \frac{1349}{8112}g_0 + \frac{25}{26}g_4 + \frac{341}{416}g_6 + \frac{1}{416}g_7 \quad (\text{B.15})$$

$$\mathcal{E}_{\text{curv}} = \frac{359}{5211}g_0 + \frac{80}{193}g_4 + \frac{613}{3474}g_5 + \frac{47}{5211}g_7 \quad (\text{B.16})$$

B.3 Case A: the main full region

Here $c \geq 0$, $\mathbf{p}_0 \geq \mathbf{p}_b$, and the long-direction powers are used:

$$g_7 = c, \quad g_8 = \mathbf{p}_0 - \mathbf{p}_b, \quad \mathbf{p} = \mathbf{p}_0, \quad \mathbf{a} = \frac{\mathbf{p} - h - v}{2}.$$

The 15 certificates are

$$\mathcal{D} = \frac{7}{54}g_2 + \frac{8}{63}g_3 + \frac{59}{42}g_6 \quad (\text{B.17})$$

$$\mathcal{C}_1 = \frac{7}{54}g_2 + \frac{8}{63}g_3 + \frac{59}{42}g_6 \quad (\text{B.18})$$

$$\mathcal{C}_2 = \frac{5}{19344}g_0 + \frac{59}{432}g_2 + \frac{2}{63}g_3 + \frac{929}{672}g_6 \quad (\text{B.19})$$

$$\mathcal{C}_3 = \frac{35849}{1695018}g_0 + \frac{15}{4907}g_4 + \frac{42979}{39256}g_6 + \frac{7}{1402}g_8 \quad (\text{B.20})$$

$$\mathcal{C}_4 = \frac{10333}{437658}g_0 + \frac{43}{543}g_2 + \frac{827}{724}g_6 + \frac{3}{181}g_8 \quad (\text{B.21})$$

$$\mathcal{C}_5 = \frac{17}{3720}g_0 + \frac{29}{70}g_3 + \frac{143}{112}g_6 + \frac{1}{720}g_7 \quad (\text{B.22})$$

$$\mathcal{C}_6 = \frac{41099}{1695018}g_0 + \frac{260}{701}g_4 + \frac{2123}{2804}g_6 + \frac{63}{2804}g_8 \quad (\text{B.23})$$

$$\mathcal{C}_7 = \frac{1}{8}g_8 \quad (\text{B.24})$$

$$\mathbf{p} - \frac{1}{1000} = \frac{4007}{93000}g_0 + \frac{39}{35}g_3 + \frac{13}{28}g_6 + \frac{11}{1260}g_7 \quad (\text{B.25})$$

$$-\mathbf{a} - \frac{1}{1000} = \frac{1963783}{15717000}g_0 + \frac{20}{91}g_4 + \frac{11}{364}g_6 + \frac{19}{3276}g_7 \quad (\text{B.26})$$

$$f + h - u - \frac{1}{1000} = \frac{179623}{3627000}g_0 + \frac{47}{180}g_2 + \frac{1}{3}g_4 + g_5 \quad (\text{B.27})$$

$$v - h - \frac{1}{1000} = \frac{9791}{1209000}g_0 + \frac{1}{3}g_2 + g_3 + g_6 \quad (\text{B.28})$$

$$\mathcal{E}_{\text{end}} = \frac{130126}{847509}g_0 + \frac{520}{701}g_4 + \frac{721}{1402}g_6 + \frac{63}{1402}g_8 \quad (\text{B.29})$$

$$\mathcal{E}_{\text{curv}} = \frac{166567}{6586632}g_0 + \frac{22193}{81720}g_2 + \frac{319}{1362}g_4 + \frac{21}{454}g_8 \quad (\text{B.30})$$

$$\mathcal{Q}_1 = \frac{1}{126}g_7 \quad (\text{B.31})$$

The first two equalities are the active balances. In particular,

$$\mathcal{D} = \mathcal{C}_1 = \frac{7}{54}(\rho - u) + \frac{8}{63}(v - u/3) + \frac{59}{42}(h_{\text{cap}} - h).$$

The second correlation has the strict margin

$$\mathcal{C}_2 = \frac{5}{19344} + \frac{59}{432}(\rho - u) + \frac{2}{63}(v - u/3) + \frac{929}{672}(h_{\text{cap}} - h).$$

B.4 Case A: the boundary full region

Here $c \geq 0$, $\mathfrak{p}_b \geq \mathfrak{p}_0$, and the conservative powers from (4.4) must be used:

$$g_7 = c, \quad g_8 = \mathfrak{p}_b - \mathfrak{p}_0, \quad \mathfrak{p} = \mathfrak{p}_b, \quad \mathfrak{a} = \frac{\mathfrak{p} - h - v}{2}.$$

The last correlation deficit is zero, as expected from $\Pi = \Pi_b$. The 15 certificates are

$$\mathcal{D} = \frac{640}{72943}g_0 + \frac{187}{1086}g_2 + \frac{595}{362}g_6 + \frac{189}{362}g_8 \quad (\text{B.32})$$

$$\mathcal{C}_1 = \frac{149}{241800}g_0 + \frac{1331}{3000}g_2 + \frac{33}{50}g_4 + \frac{277}{100}g_6 \quad (\text{B.33})$$

$$\mathcal{C}_2 = \frac{15329}{13540800}g_0 + \frac{72251}{168000}g_2 + \frac{1593}{2800}g_4 + \frac{3673}{1400}g_6 \quad (\text{B.34})$$

$$\mathcal{C}_3 = \frac{23}{1488}g_0 + \frac{27}{80}g_2 + \frac{3}{4}g_4 + \frac{5}{2}g_6 \quad (\text{B.35})$$

$$\mathcal{C}_4 = \frac{1771}{80600}g_0 + \frac{847}{3000}g_2 + \frac{21}{50}g_4 + \frac{199}{100}g_6 \quad (\text{B.36})$$

$$\mathcal{C}_5 = \frac{41}{1209}g_0 + \frac{8}{15}g_2 + g_4 + \frac{31}{8}g_6 \quad (\text{B.37})$$

$$\mathcal{C}_6 = \frac{75}{3224}g_0 + \frac{7}{120}g_2 + \frac{1}{2}g_4 + g_6 \quad (\text{B.38})$$

$$\mathcal{C}_7 = 0g_0 \quad (\text{B.39})$$

$$\mathfrak{p} - \frac{1}{1000} = \frac{12582057}{72943000}g_0 + \frac{7}{543}g_2 + \frac{785}{362}g_6 + \frac{441}{362}g_8 \quad (\text{B.40})$$

$$-\mathfrak{a} - \frac{1}{1000} = \frac{55847}{403000}g_0 + \frac{61}{60}g_2 + 3g_4 + 5g_6 \quad (\text{B.41})$$

$$f + h - u - \frac{1}{1000} = \frac{82328171}{823329000}g_0 + \frac{4907}{20430}g_2 + \frac{362}{681}g_4 + \frac{21}{227}g_8 \quad (\text{B.42})$$

$$v - h - \frac{1}{1000} = \frac{16892171}{218829000}g_0 + \frac{121}{181}g_2 + \frac{1043}{362}g_6 + \frac{63}{362}g_8 \quad (\text{B.43})$$

$$\mathcal{E}_{\text{end}} = \frac{733}{4836}g_0 + \frac{7}{60}g_2 + g_4 + g_6 \quad (\text{B.44})$$

$$\mathcal{E}_{\text{curv}} = \frac{75}{3224}g_0 + \frac{47}{120}g_2 + \frac{1}{2}g_4 + \frac{1}{2}g_6 \quad (\text{B.45})$$

$$\mathcal{Q}_1 = \frac{11899}{282503}g_0 + \frac{3560}{4907}g_4 + \frac{9897}{9814}g_6 + \frac{909}{1402}g_8 \quad (\text{B.46})$$

B.5 Case B

On $u_0 \leq u \leq \rho$, $9u/20 \leq v \leq u$, take

$$b_0 = 1, \quad b_1 = u - u_0, \quad b_2 = \rho - u, \quad b_3 = v - 9u/20, \quad b_4 = u - v.$$

For the seven exponents in (7.13), the certificates are

$$\theta - B_1 = \frac{415}{9672}b_0 + \frac{13}{120}b_2 + \frac{1}{2}b_3 \quad (\text{B.47})$$

$$\theta - B_2 = \frac{179}{24180}b_0 + \frac{7}{40}b_1 + \frac{3}{2}b_3 \quad (\text{B.48})$$

$$\theta - B_3 = \frac{11}{241800}b_0 + \frac{3}{400}b_1 + \frac{7}{20}b_3 \quad (\text{B.49})$$

$$\theta - B_4 = \frac{1699}{106392}b_0 + \frac{3}{88}b_2 + \frac{5}{22}b_3 \quad (\text{B.50})$$

$$\theta - B_5 = \frac{593}{24180}b_0 + \frac{9}{40}b_1 + \frac{1}{2}b_3 \quad (\text{B.51})$$

$$\theta - B_6 = \frac{69}{403}b_0 + \frac{1}{6}b_2 \quad (\text{B.52})$$

$$\theta - B_7 = \frac{621}{4030}b_0 + \frac{7}{60}b_1 + b_3 \quad (\text{B.53})$$

There are $14 + 15 + 15 + 7 = 51$ identities. Each Case B constant term is strictly positive, and the smallest is $11/241800$. Definitions (B.1)–(B.2) connect all certificates to the chosen local monomials and to both first-B errors. Together with the domain, range and integer-rounding arguments in the main text, they complete the parameter verification.

References

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